

Robot Learning with Inspiration from Child Development

Jitendra Malik

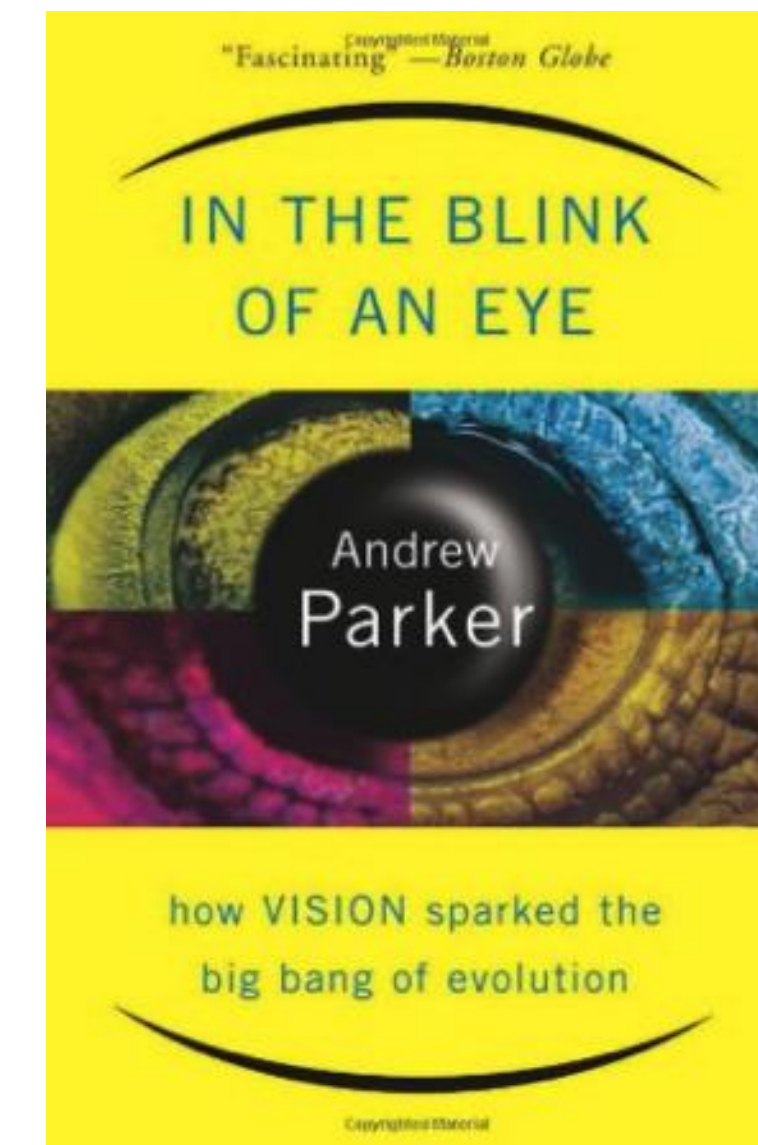
Phylogeny of Intelligence



Cambrian Explosion
540 million years ago

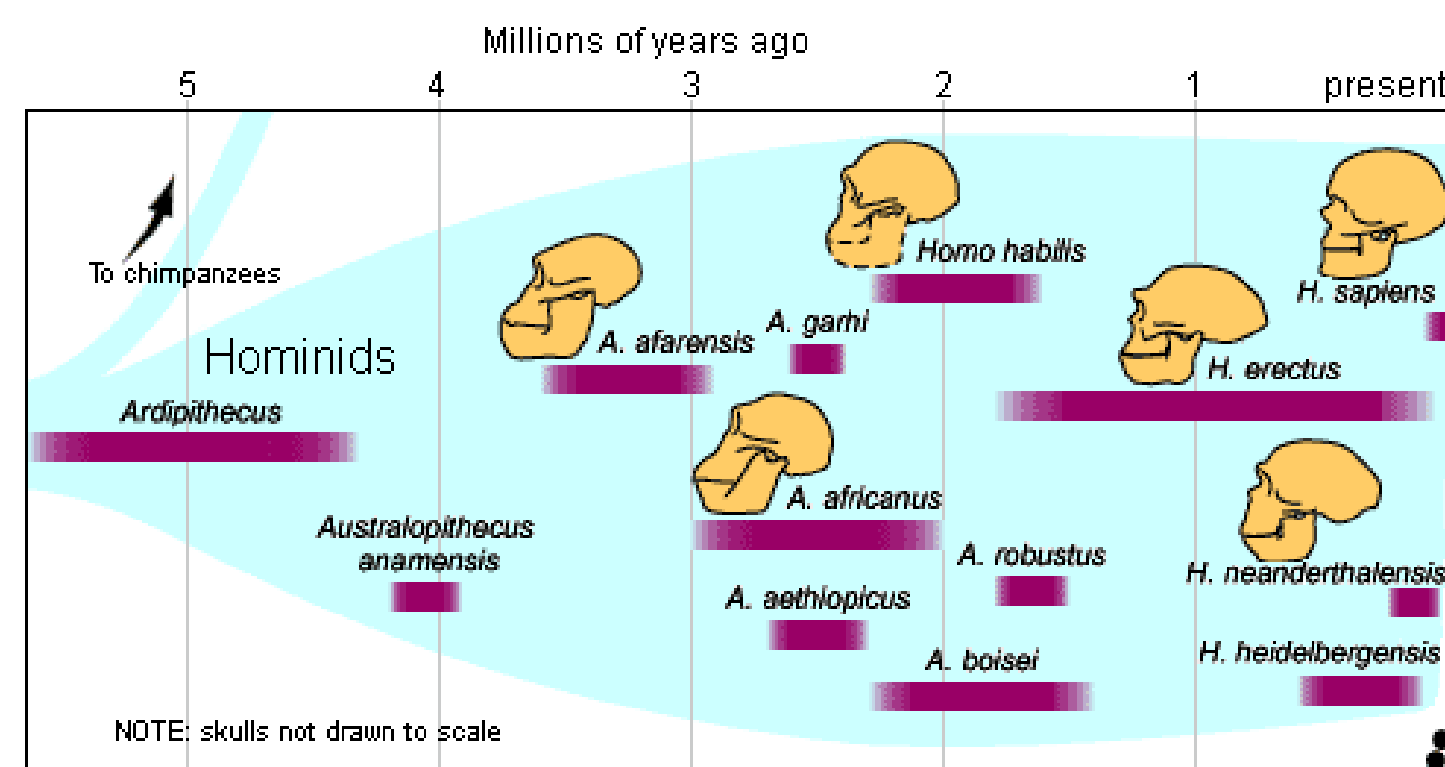
Variety of life forms,
almost all phyla emerge

Animals that could
see and move



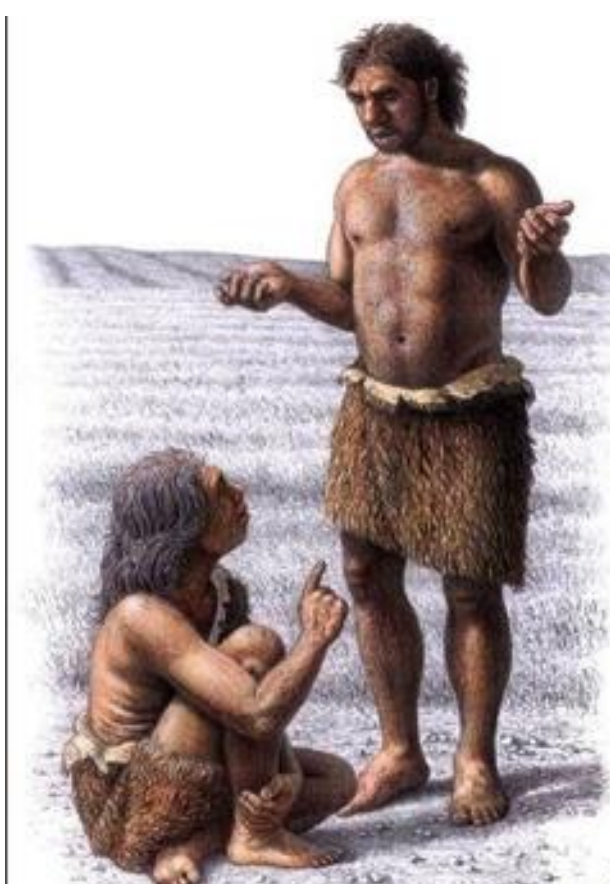
Gibson: we see in order to move and we move in order to see

Hominid evolution, last 5 million years



Bipedalism
Opposable thumb
Tool use

Modern humans, last 50 K years



Language
Abstract thinking
Symbolic behavior

Anaxagoras: It is because of his being armed with hands that man is the most intelligent animal

The Development of Embodied Cognition: Six Lessons from Babies

Linda Smith & Michael Gasser

Abstract. The embodiment hypothesis is the idea that intelligence emerges in the interaction of an agent with an environment and as a result of sensorimotor activity. In this paper we offer six lessons for *developing* embodied intelligent agents suggested by research in developmental psychology. We argue that starting as a baby grounded in a physical, social and linguistic world is crucial to the development of the flexible and inventive intelligence that characterizes humankind.

Towards an Embodied AI, inspired by child development



Perception and Interaction

456

A. M. TURING :

Instead of trying to produce a programme to simulate the adult mind, why not rather try to produce one which simulates the child's? If this were then subjected to an appropriate course of education one would obtain the adult brain. Presumably the child-brain is something like a note-book as one buys it from the stationers. Rather little mechanism, and lots of blank sheets. (Mechanism and writing are from our point of view almost synonymous.) Our hope is that there is so little mechanism in the child-brain that something like it can be easily programmed. The amount of work in the education we can assume, as a first approximation, to be much the same as for the human child.

Language

Turing (1950)
Computing Machinery
And Intelligence

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- Be multi-modal
- Be incremental
- Be physical
- Explore
- Be social
- Use language

Animal Gaits



walk



amble



pace



trot



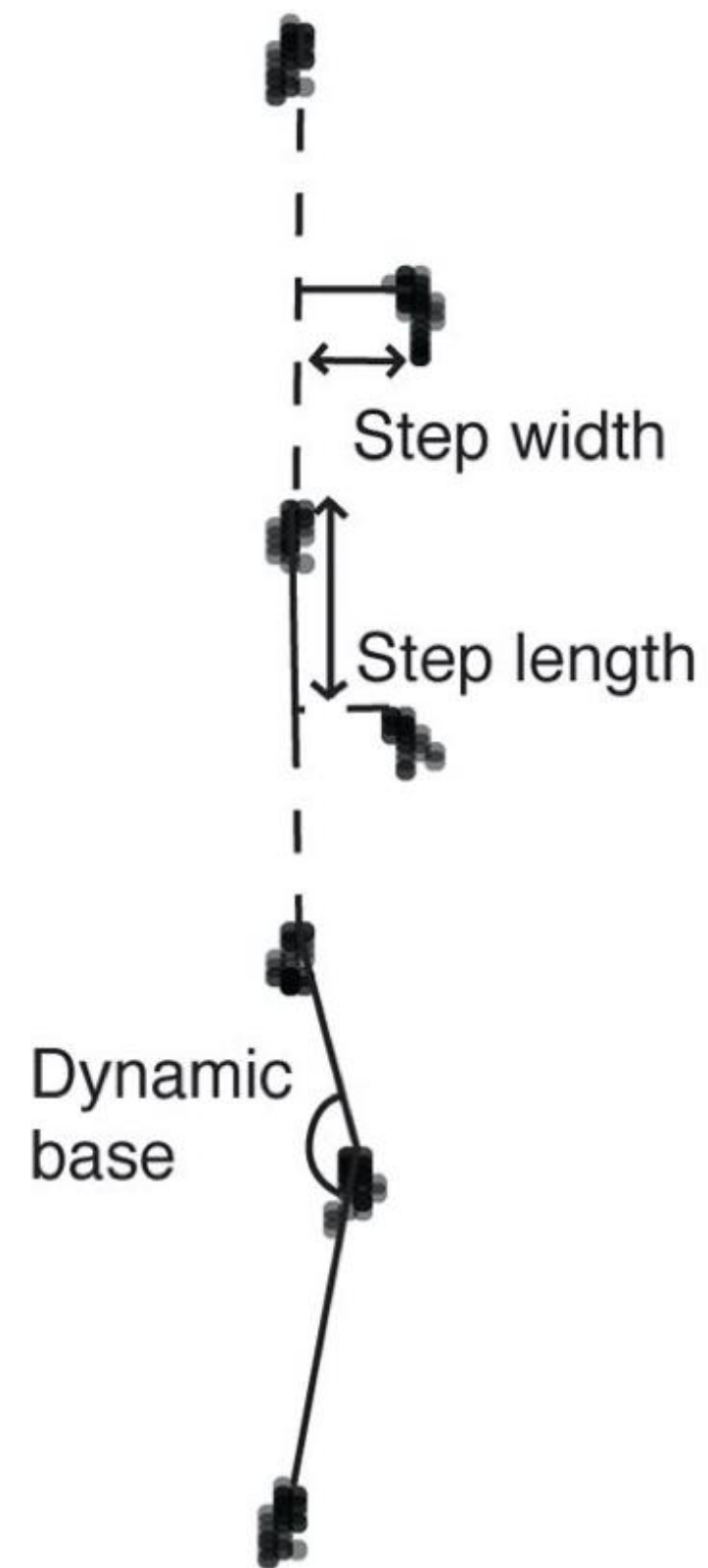
canter



gallop / run

Gaits in Real Life

A



B

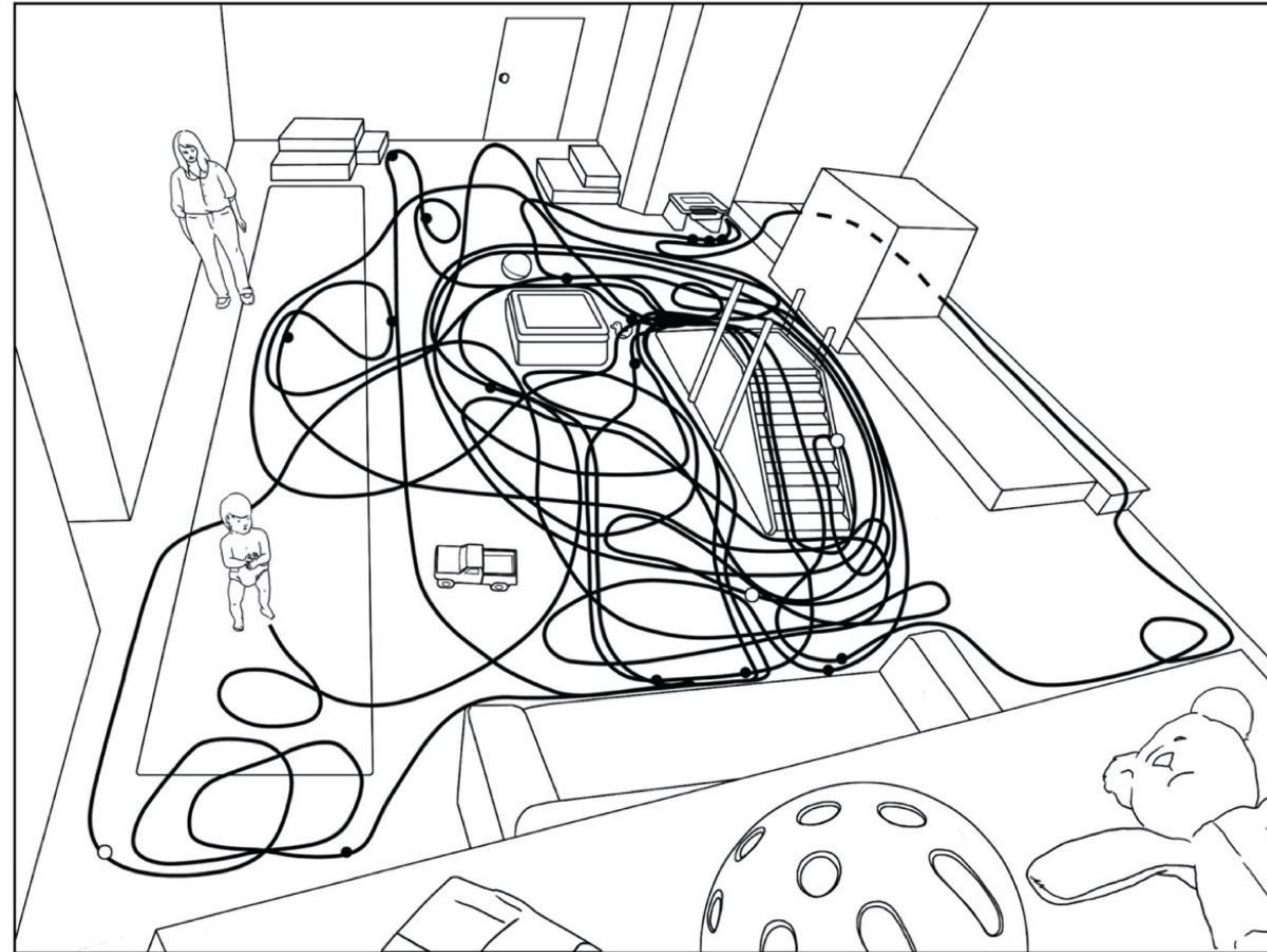


Figure 2.

(A) Sample footprint path showing the calculation of standard gait measures obtained on the pressure-sensitive gait carpet (step length, step width, and dynamic base angle). (B) Schematic drawing of the laboratory playroom. The squiggly line represents a typical path in

Cole, Whitney G., Scott R.
Robinson, and Karen E.
Adolph. "Bouts of steps: The
organization of infant
exploration." *Developmental
psychobiology* (2016)

Most walking bouts contain omnidirectional, not forward steps



Most walking bouts contain omnidirectional, not forward steps

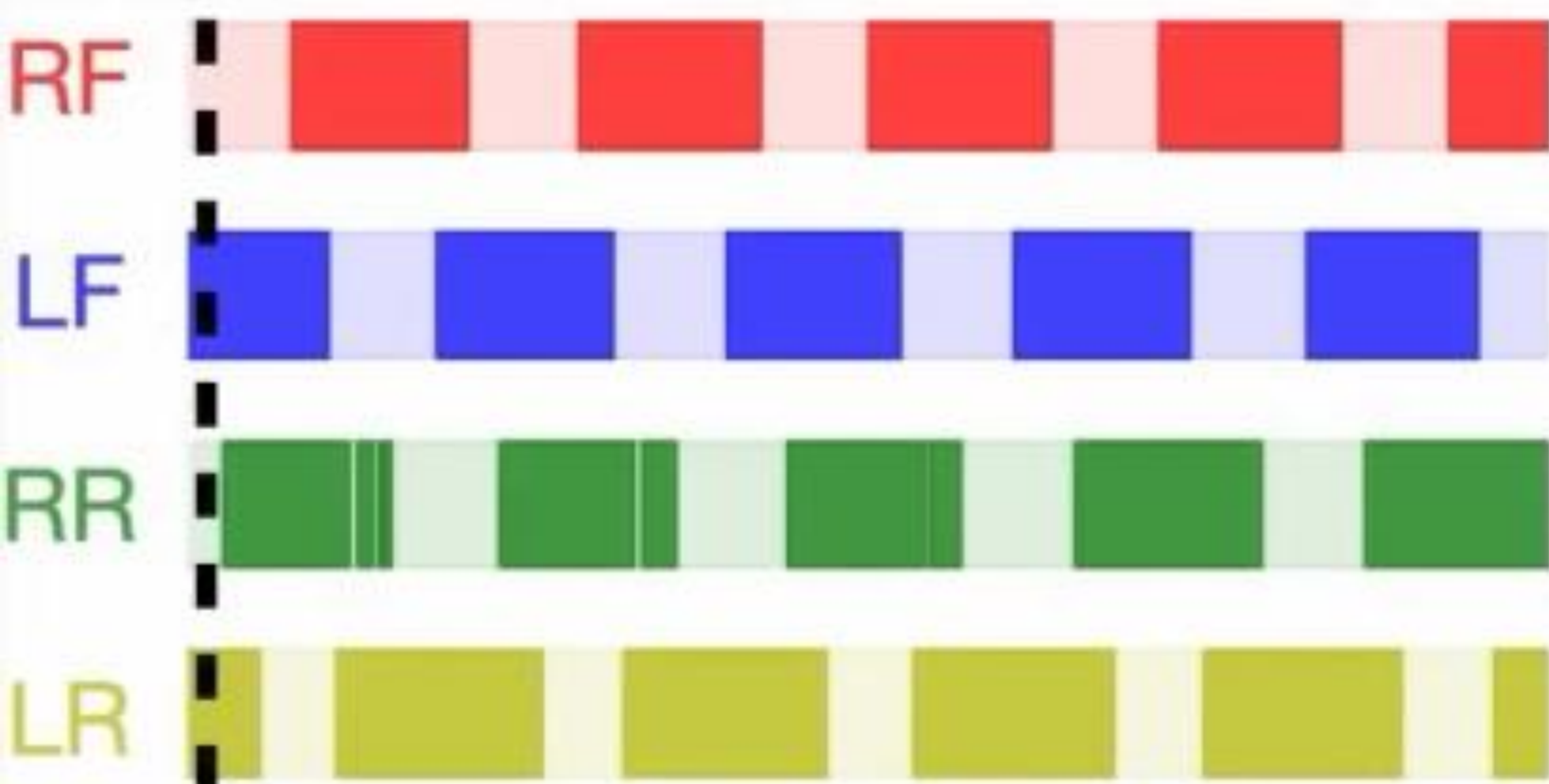


1/2 normal speed

Lee, Cole, Golenia, & Adolph (2018) *Dev Sci*

Walk

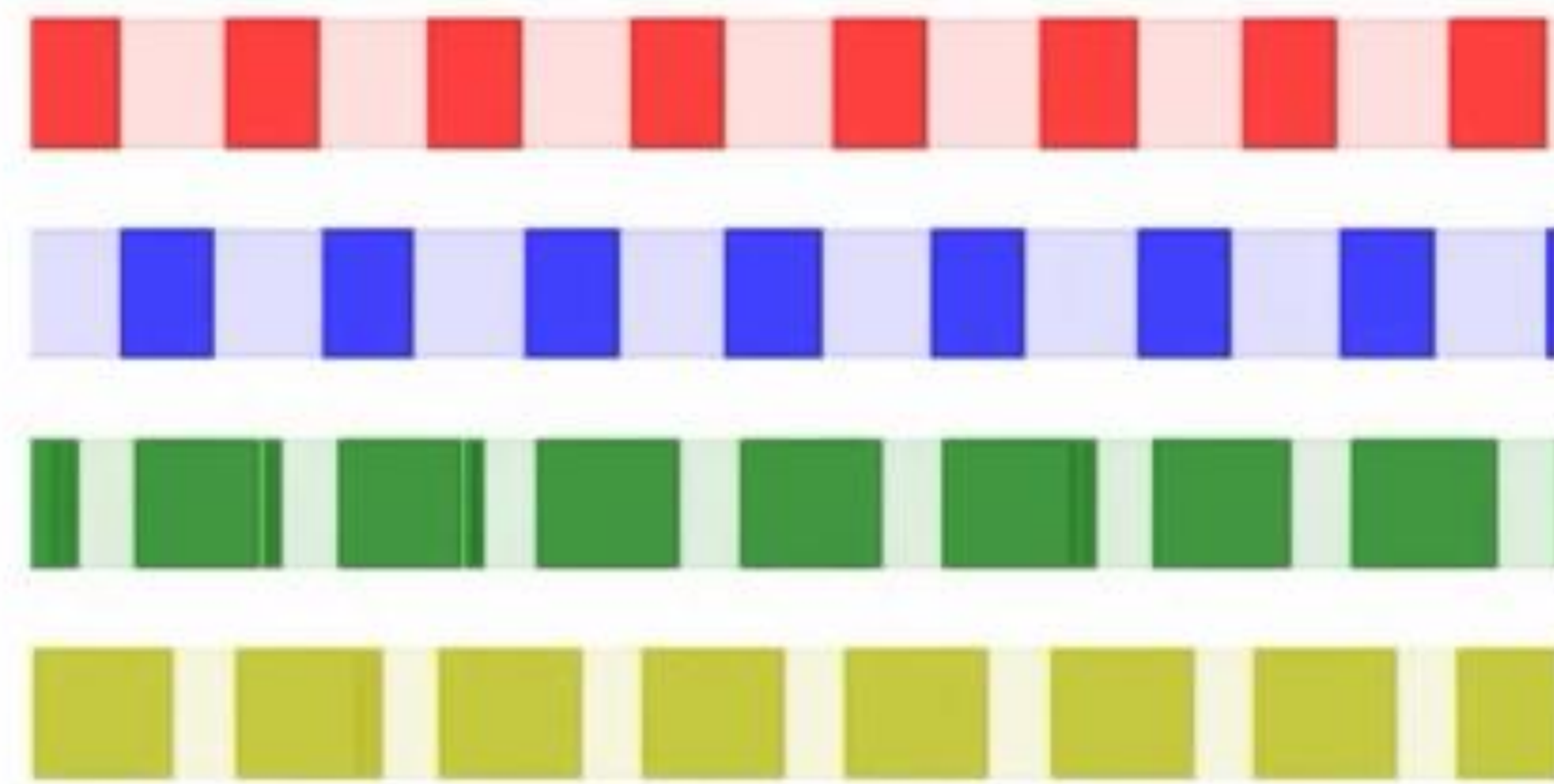
0.375 m/s



x0.5 slow

Trot

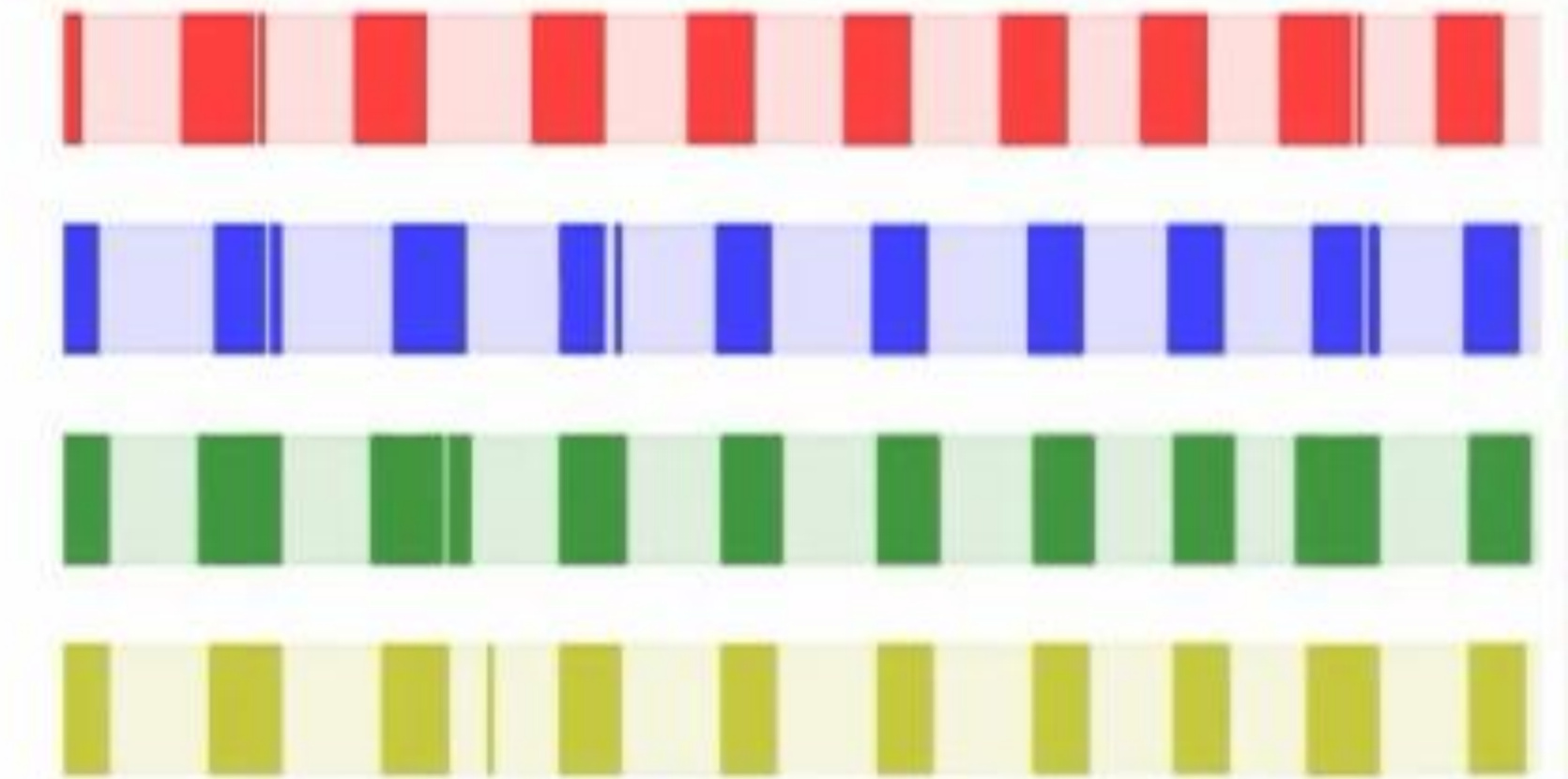
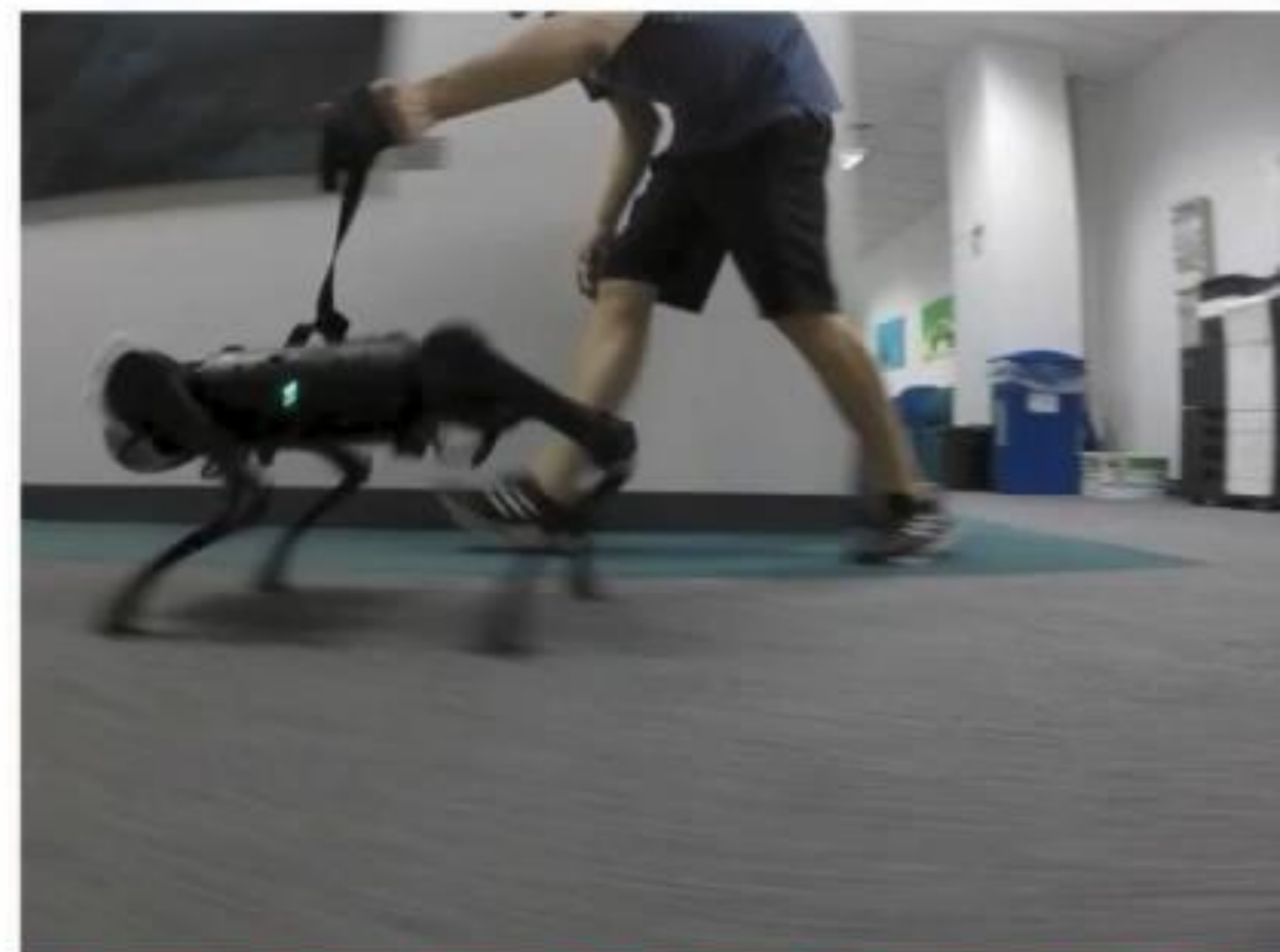
0.9 m/s



x0.25 slow

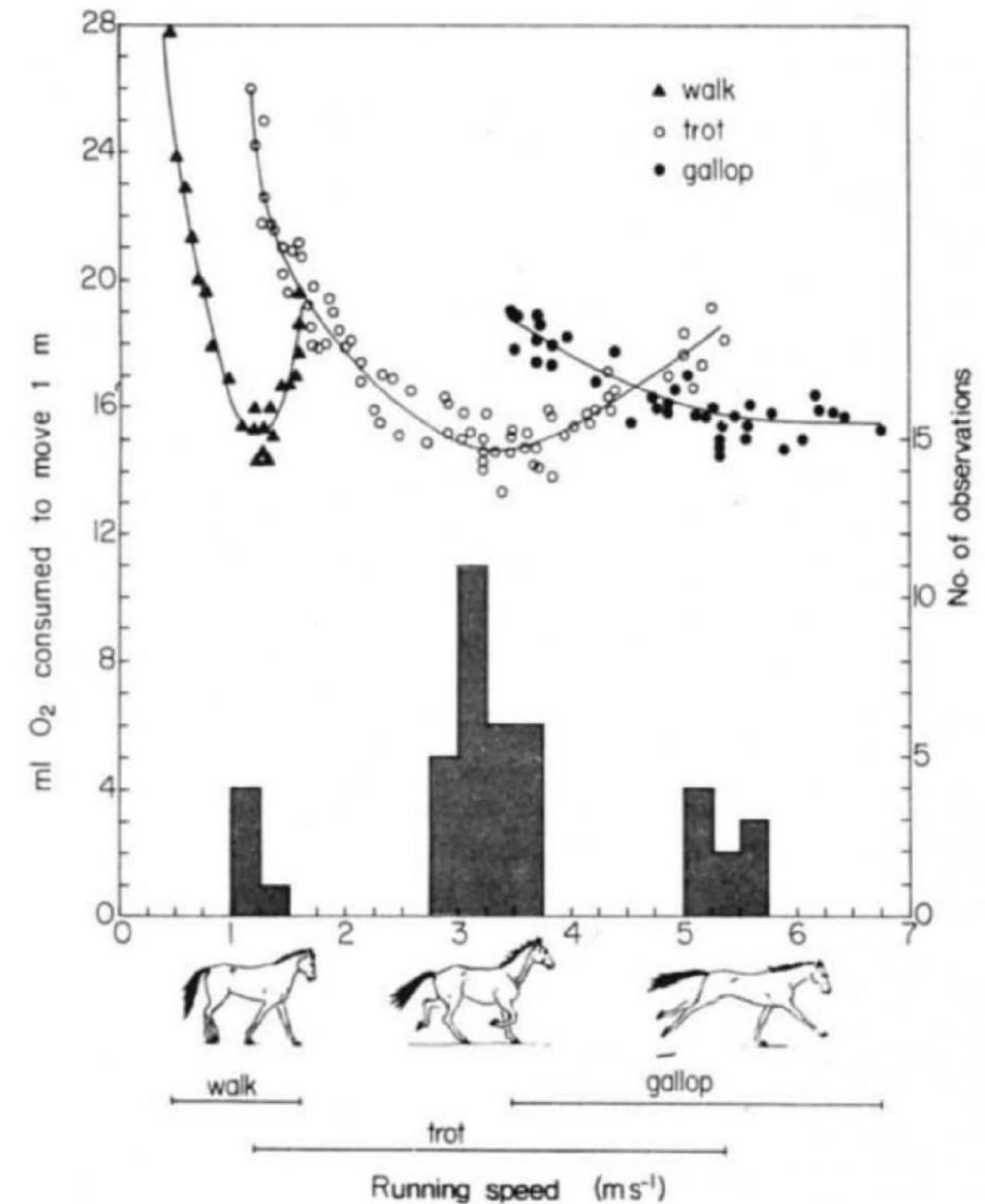
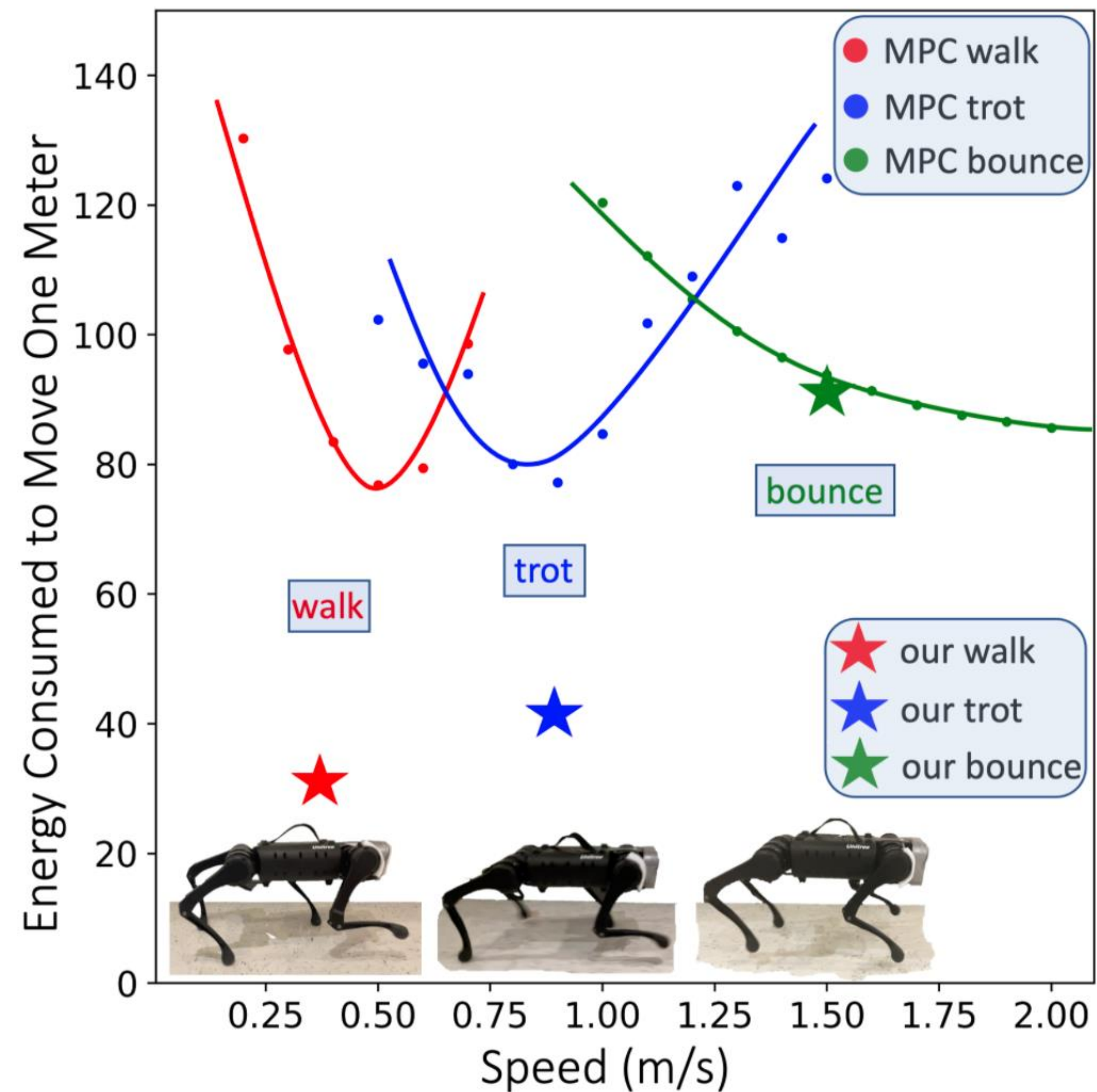
Bounce (Gallop)

1.5 m/s



x0.25 slow

Different gaits are energy efficient at different speeds



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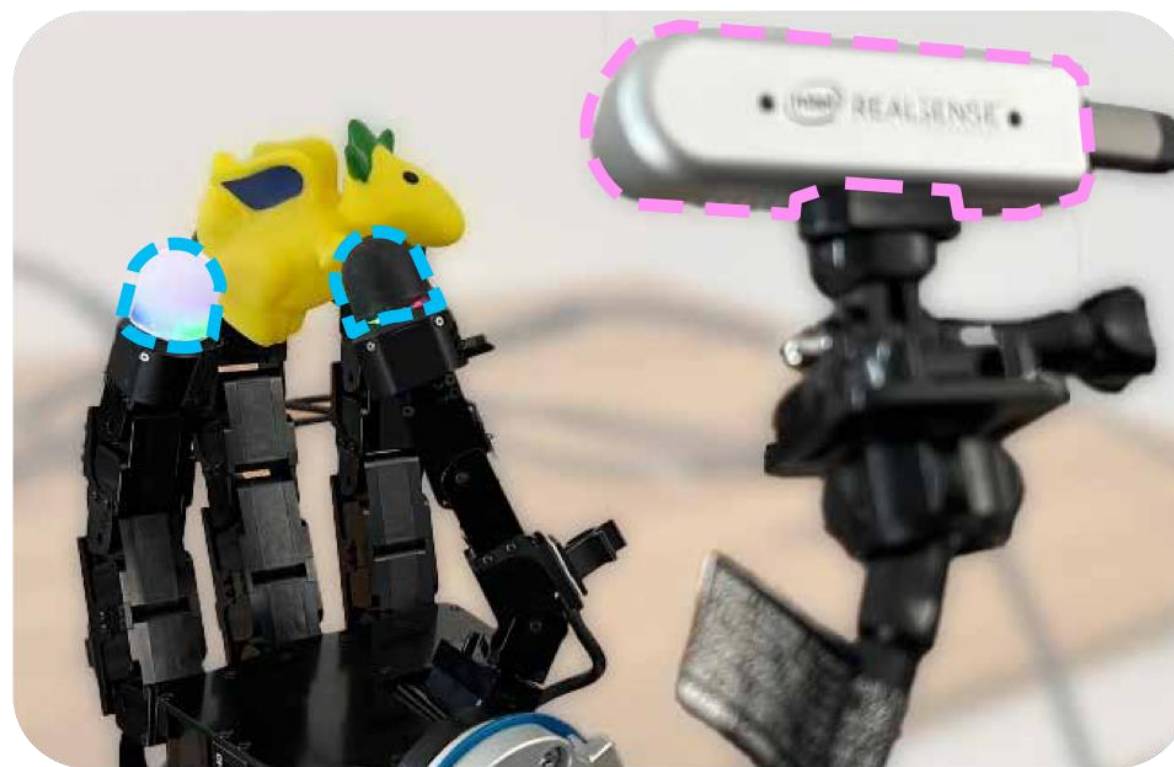
- Be multi-modal
- Be incremental
- Be physical
- Explore
- Be social
- Use language

Rotatelt: General In-Hand Object Rotation with Vision and Touch

Haozhi Qi, Brent Yi, Sudharshan Suresh, Mike Lambeta, Yi Ma, Roberto Calandra, Jitendra Malik
CORL 2023

General In-Hand Object Rotation

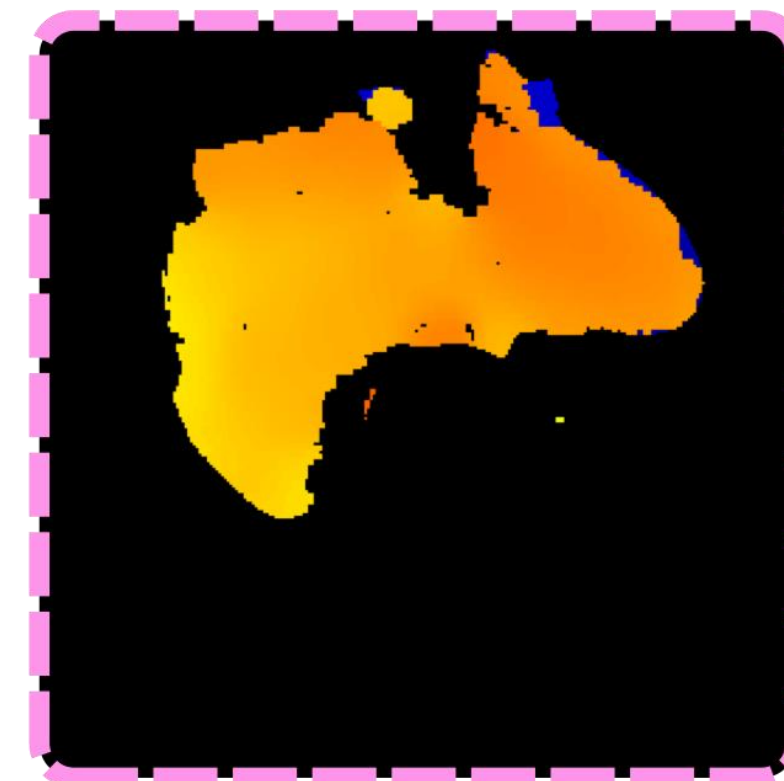
- We can train policy for any pre-specified rotation axis.
- It still generalizes to a diverse set of objects.
- It uses a combination of vision and touch, to infer the shape of the object.



Hardware Setup



Raw Depth



Object Depth



Tactile Image (x4)

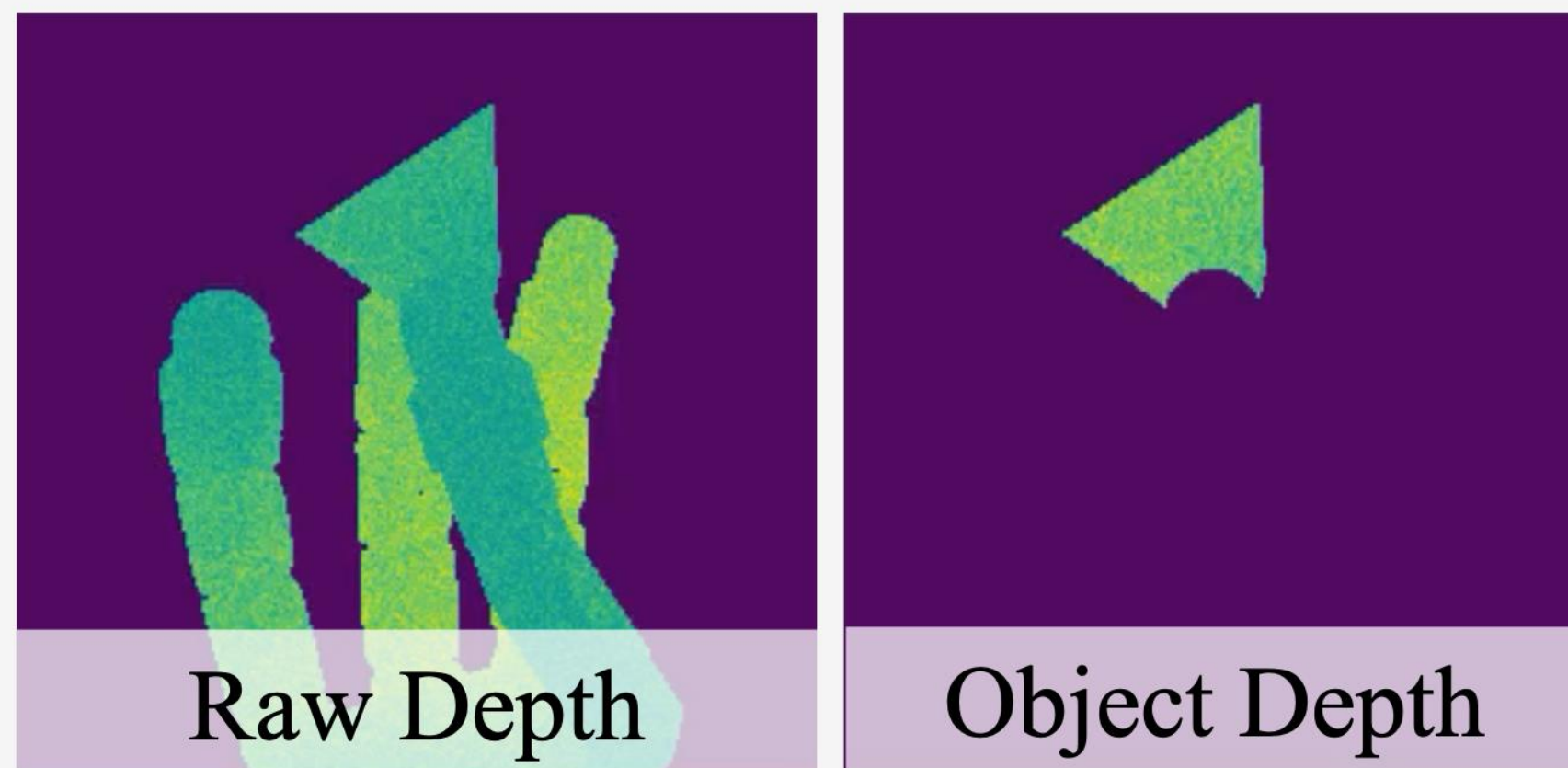
[1] Haozhi Qi, Ashish Kumar, Roberto Calandra, Yi Ma, Jitendra Malik. In-Hand Object Rotation via Rapid Motor Adaptation. CoRL 2022



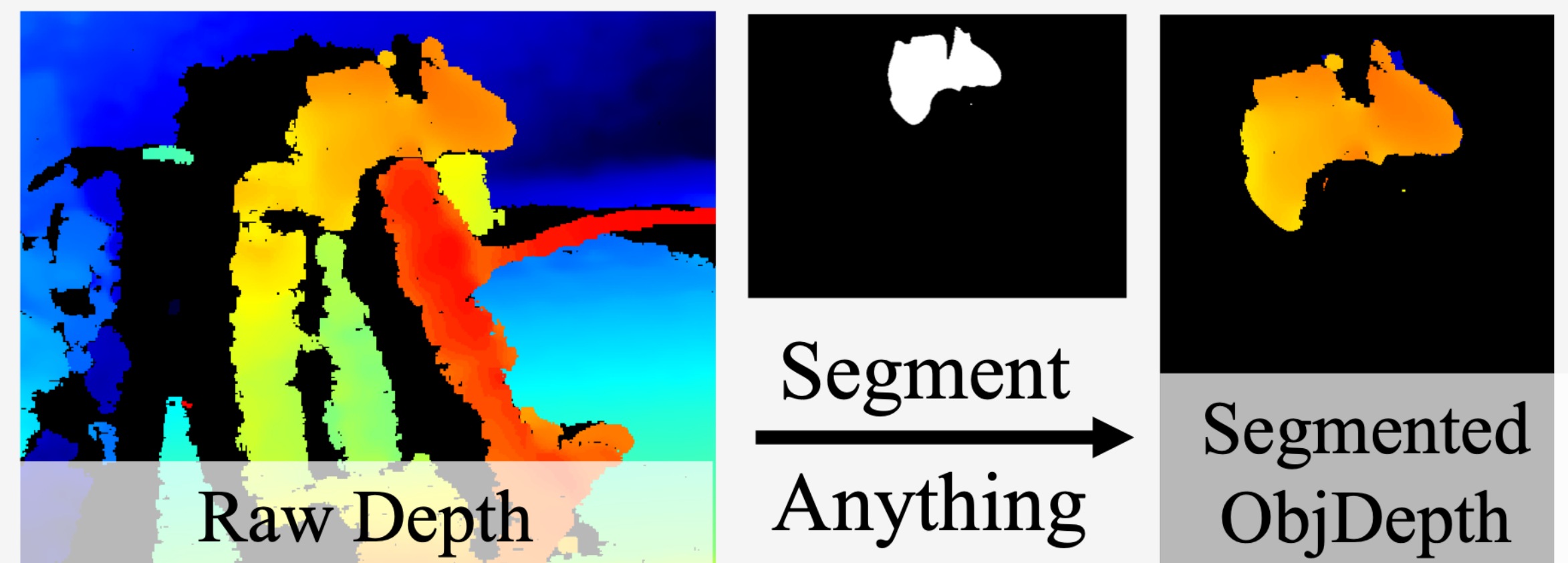
Visuotactile Policy Training: Vision

- Finding Visuotactile Information to minimize sim2real gap.
- The visuotactile information should also be easy to be simulated.
- For vision: object depth.

(a) Simulated Vision

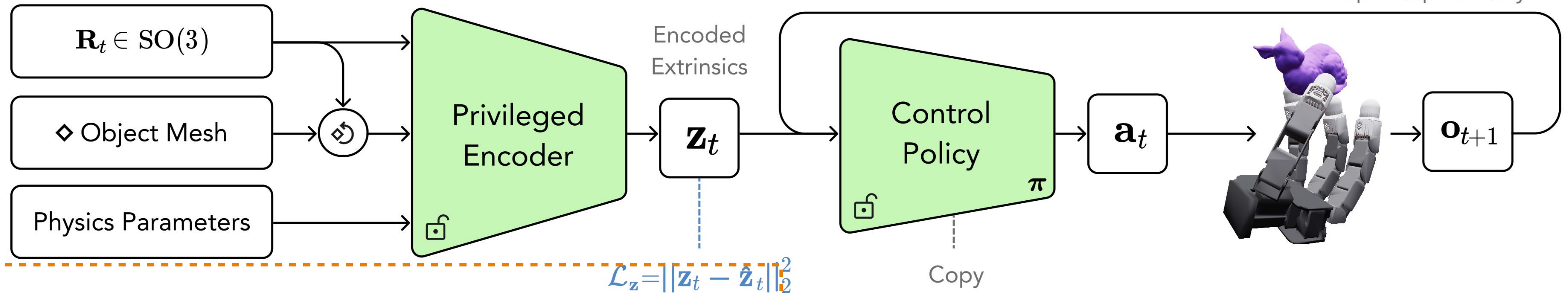


(b) Real Vision

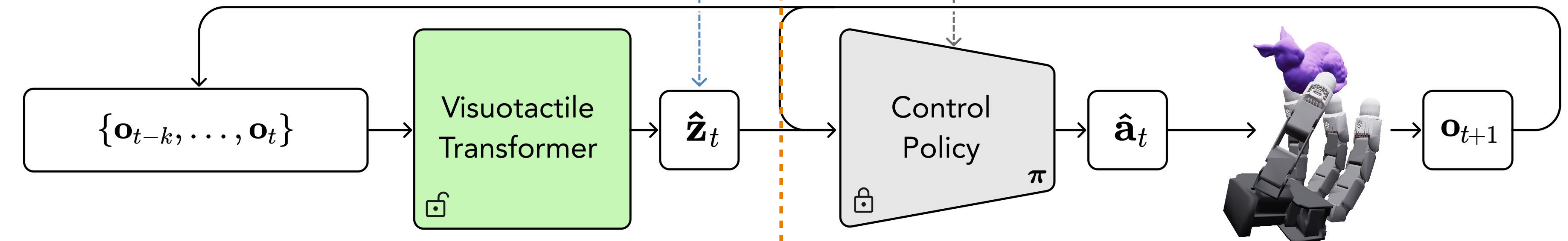


Visuotactile Policy Training

(1) Oracle Policy Training

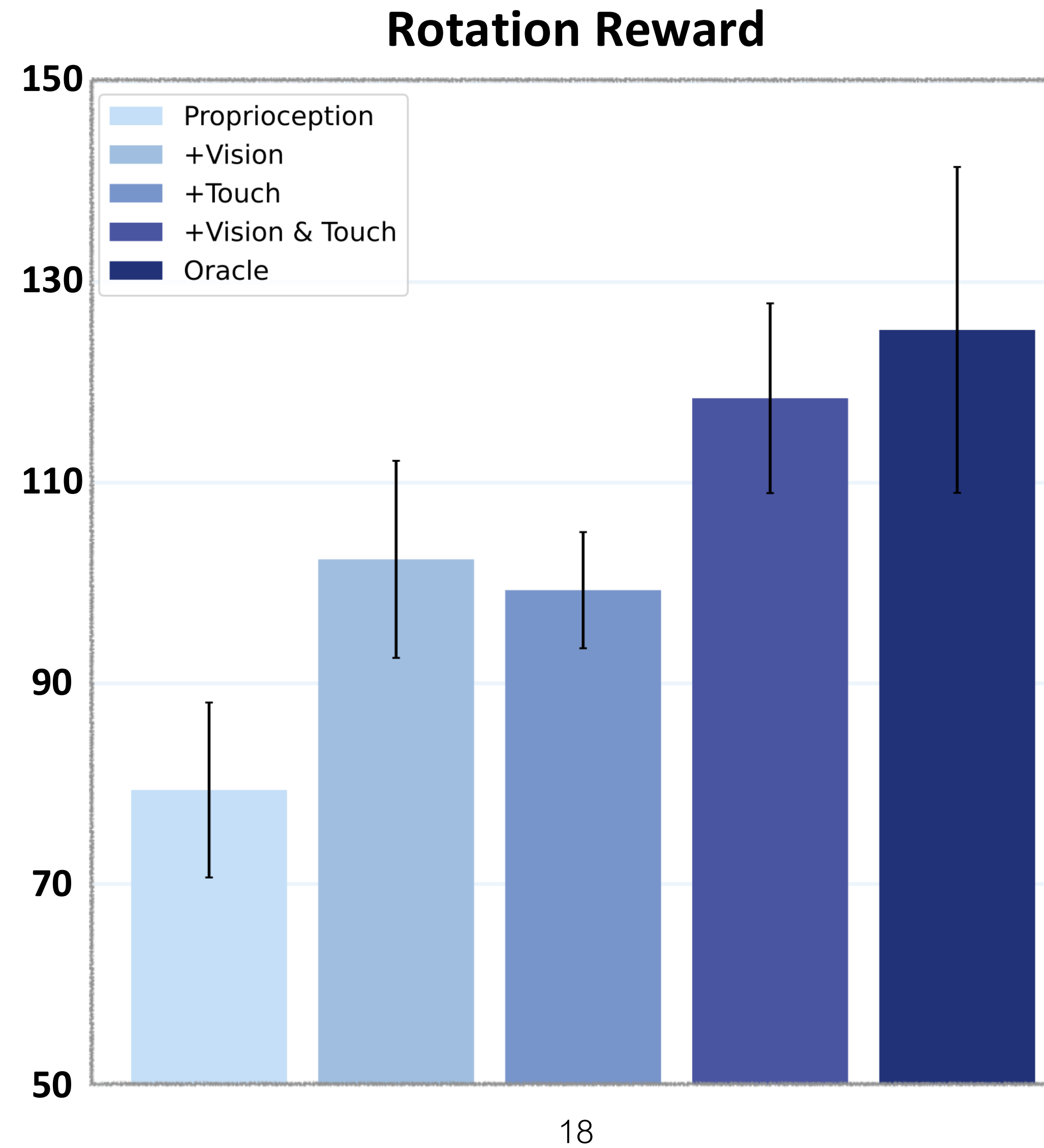


(2) Visuotactile Policy Training



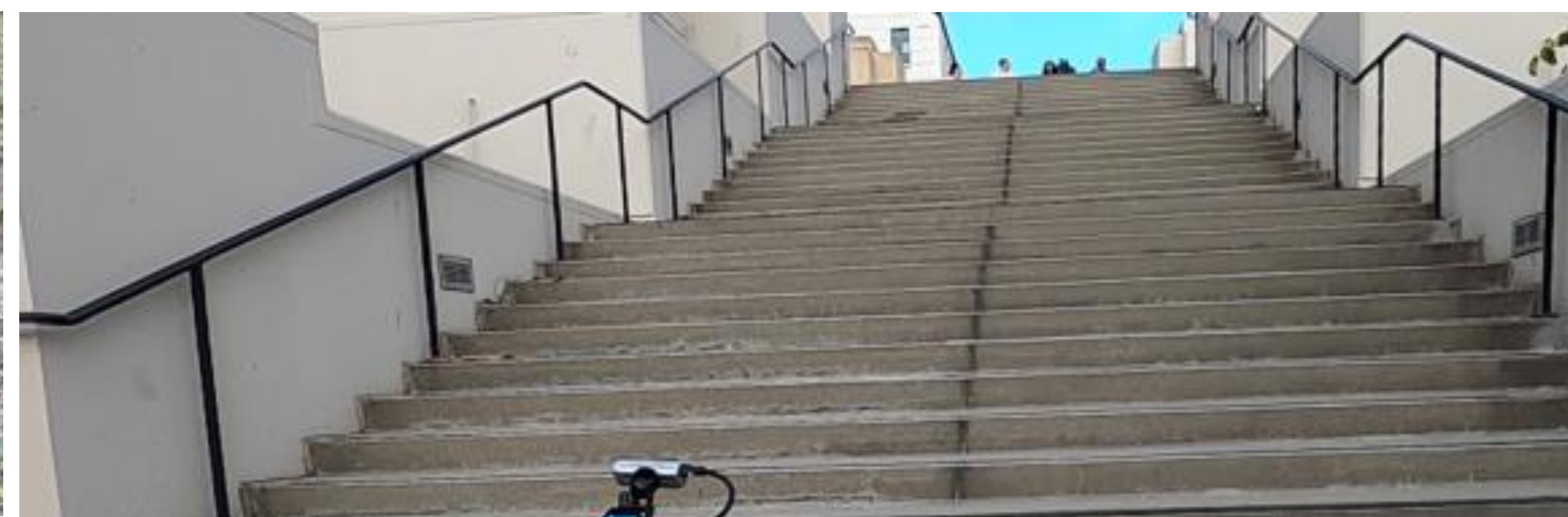
$$\mathcal{L}_z = \|\mathbf{z}_t - \hat{\mathbf{z}}_t\|_2^2$$

Proprioception + Vision + Touch

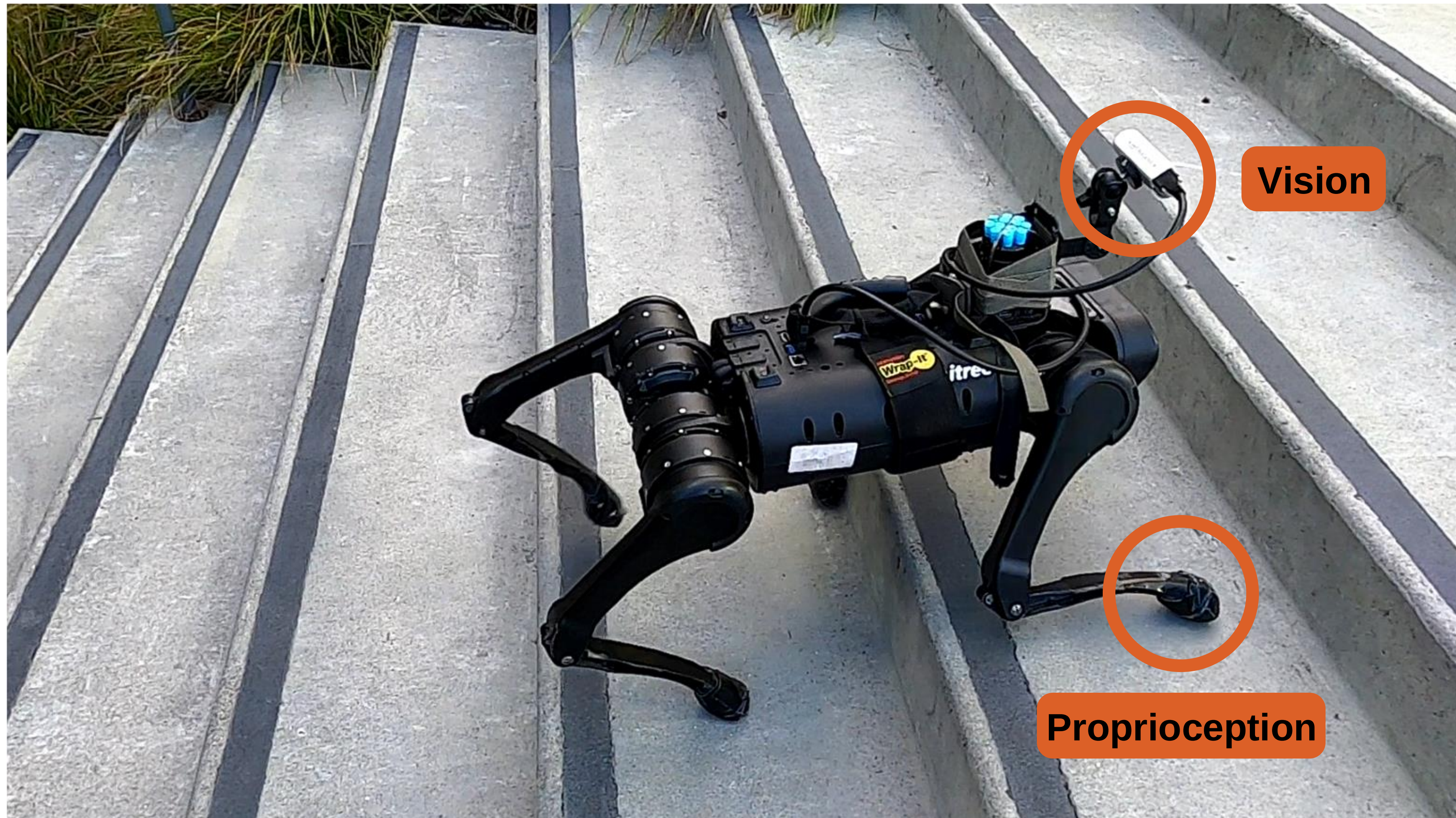


Learning Visual Locomotion with Cross-Modal Supervision

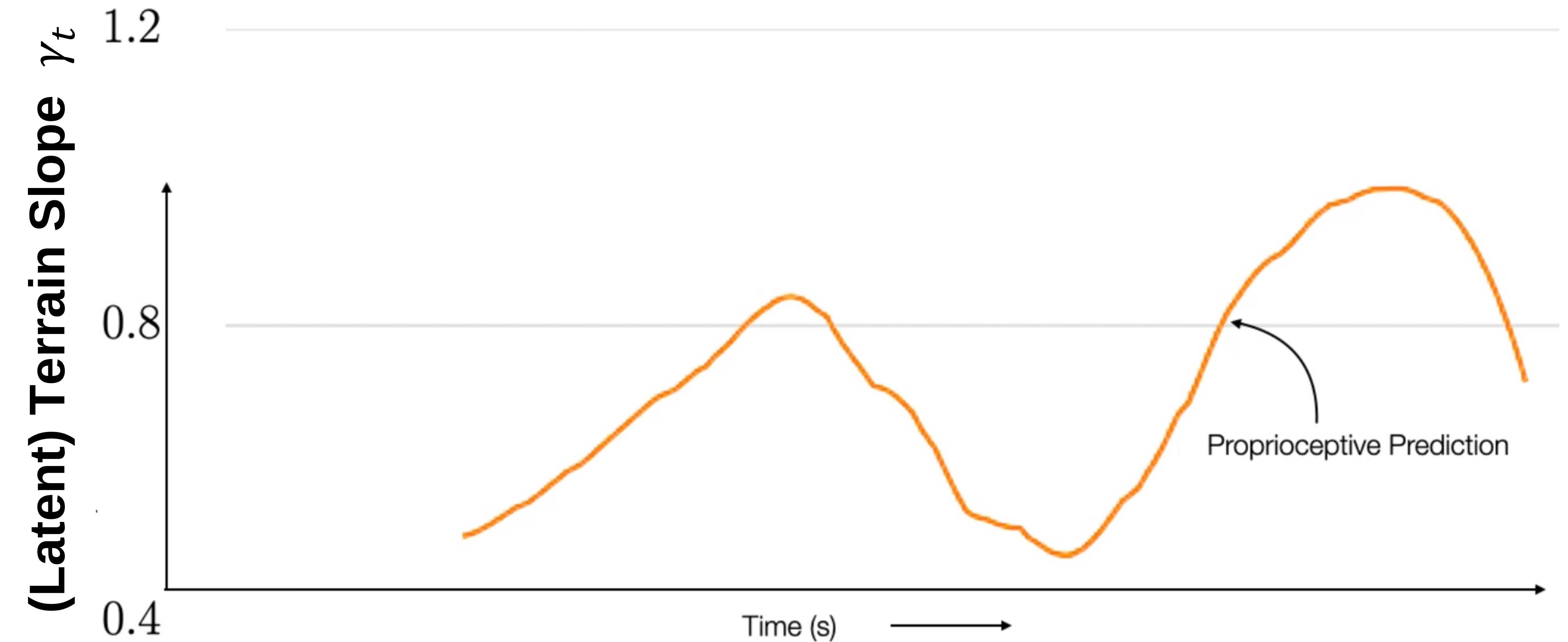
A. Loquercio, A. Kumar, J. Malik



Proprioception to Estimate Terrain Properties

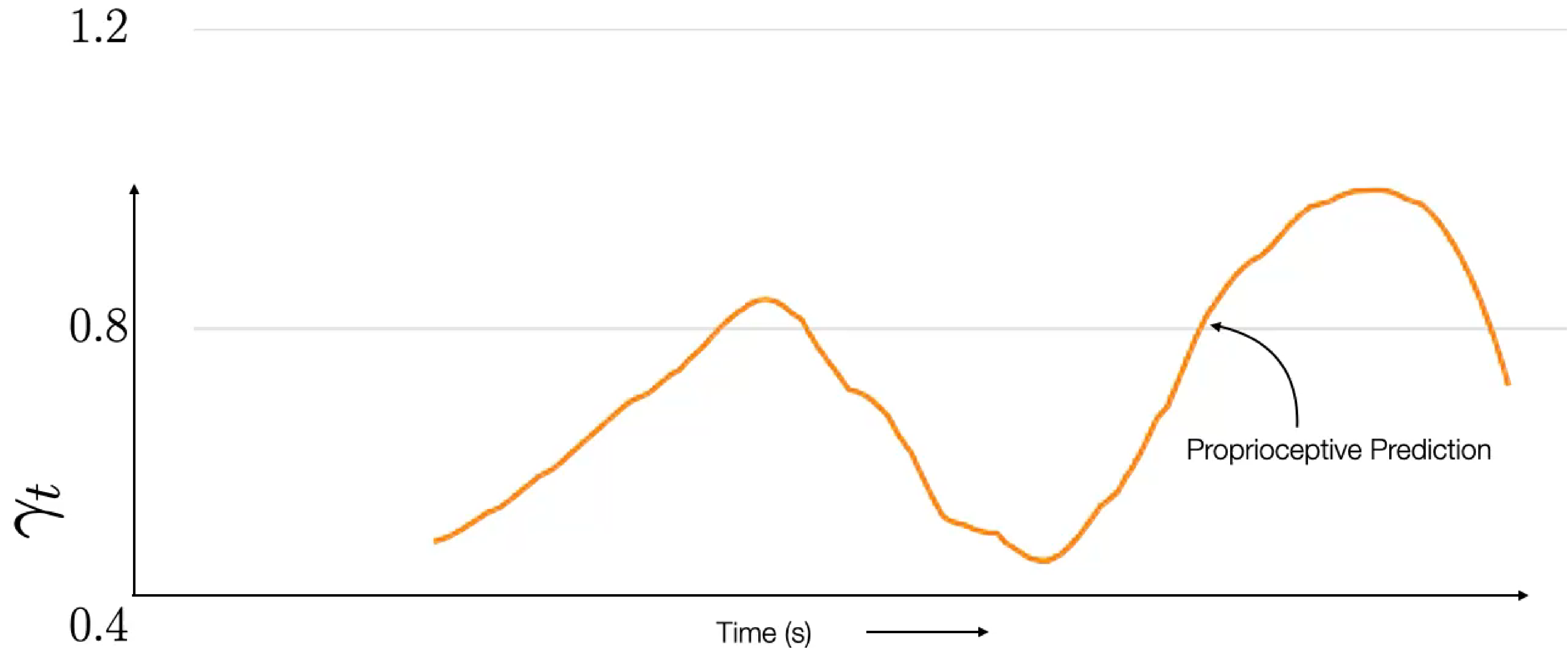


Estimate the Terrain Geometry From Proprioception



Cross-Modal Supervision: Predicting the Future Terrain Geometry from Vision

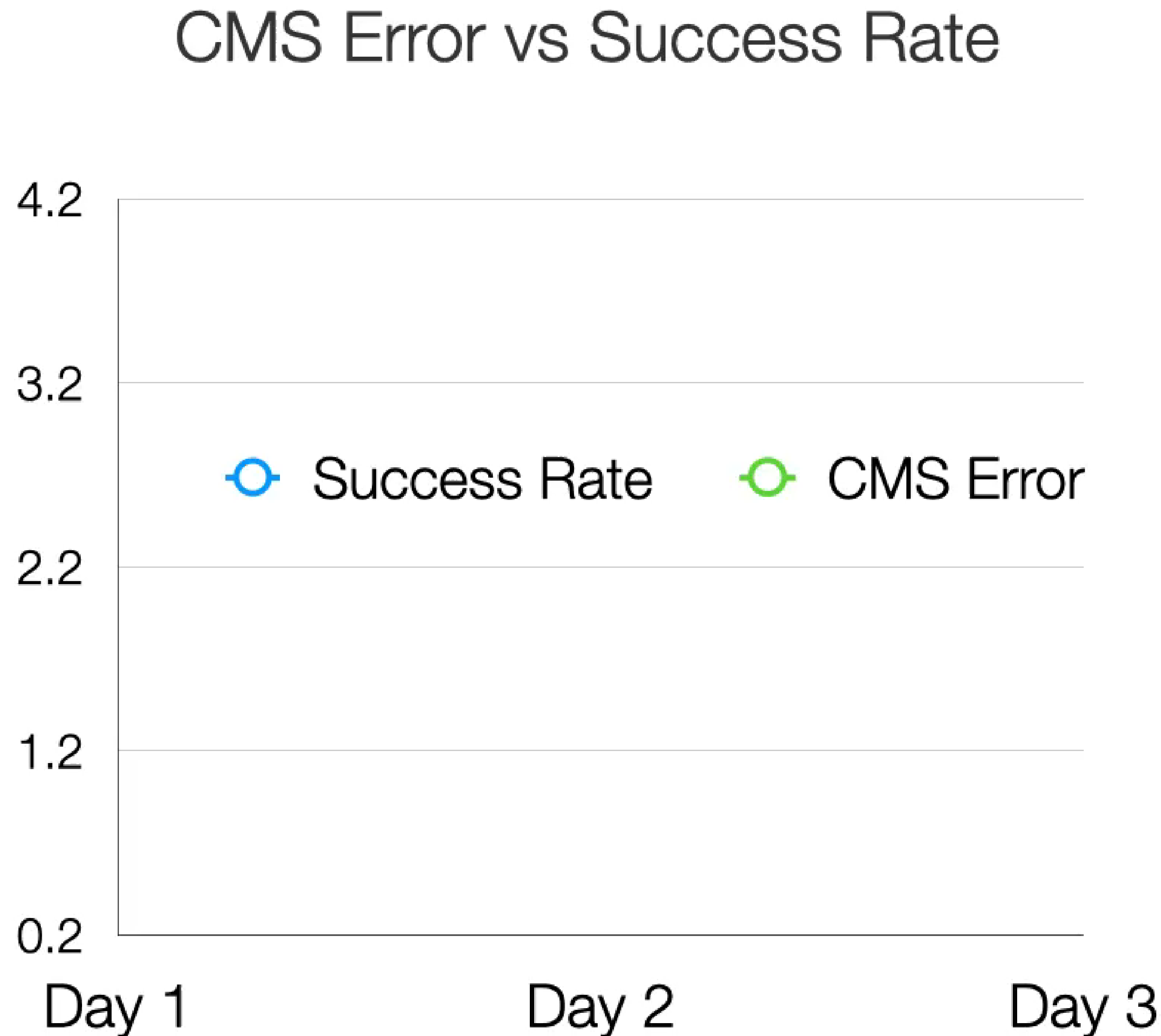
We define the difference between time-shifted proprioception and vision predictions as CMS Error



A Blind Policy To Estimate the Terrain Geometry



More Data Gives a Lower CMS Error and Better Performance

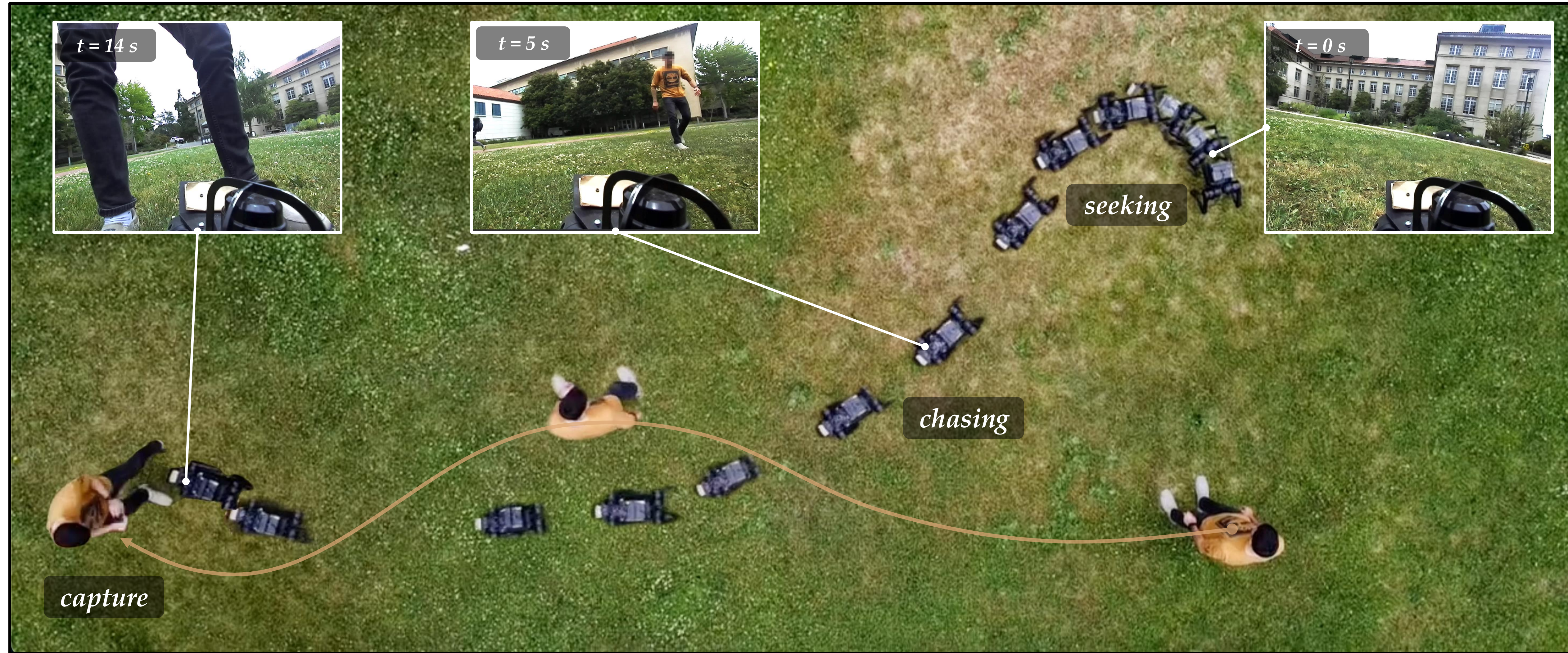
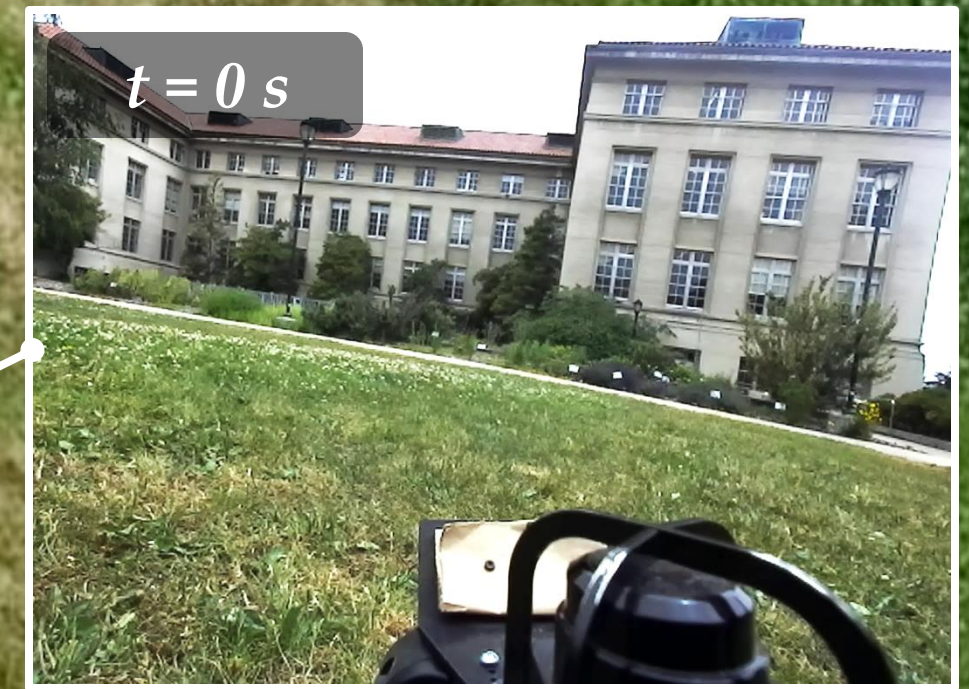


The Development of Embodied Cognition: Six Lessons from Babies

Linda Smith & Michael Gasser

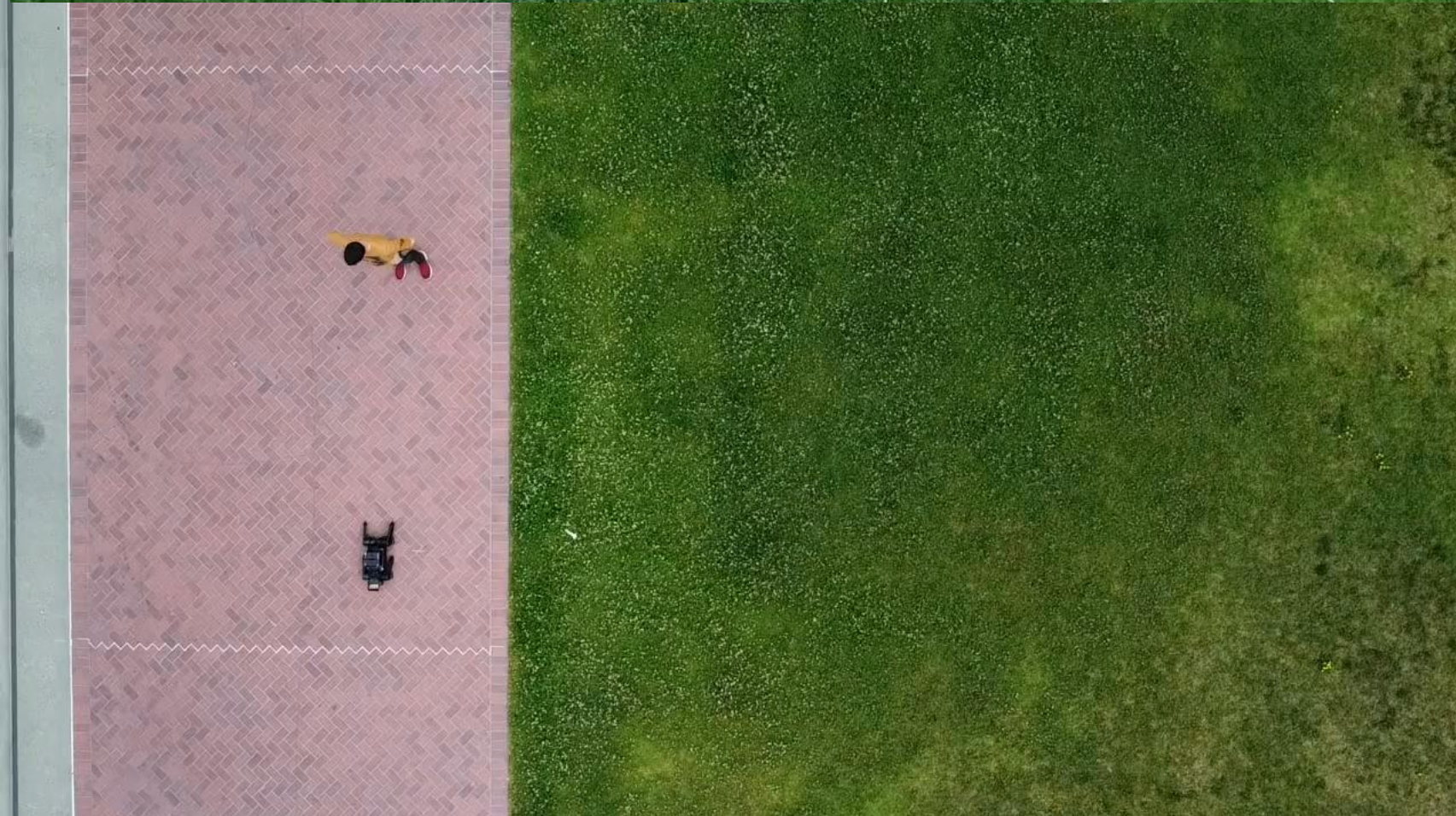
- Be multi-modal
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Learning Vision-Based Pursuit-Evasion Robot Policies



Andrea Bajcsy*, Antonio Loquercio*, Ashish Kumar, Jitendra Malik
(ICRA 2024)





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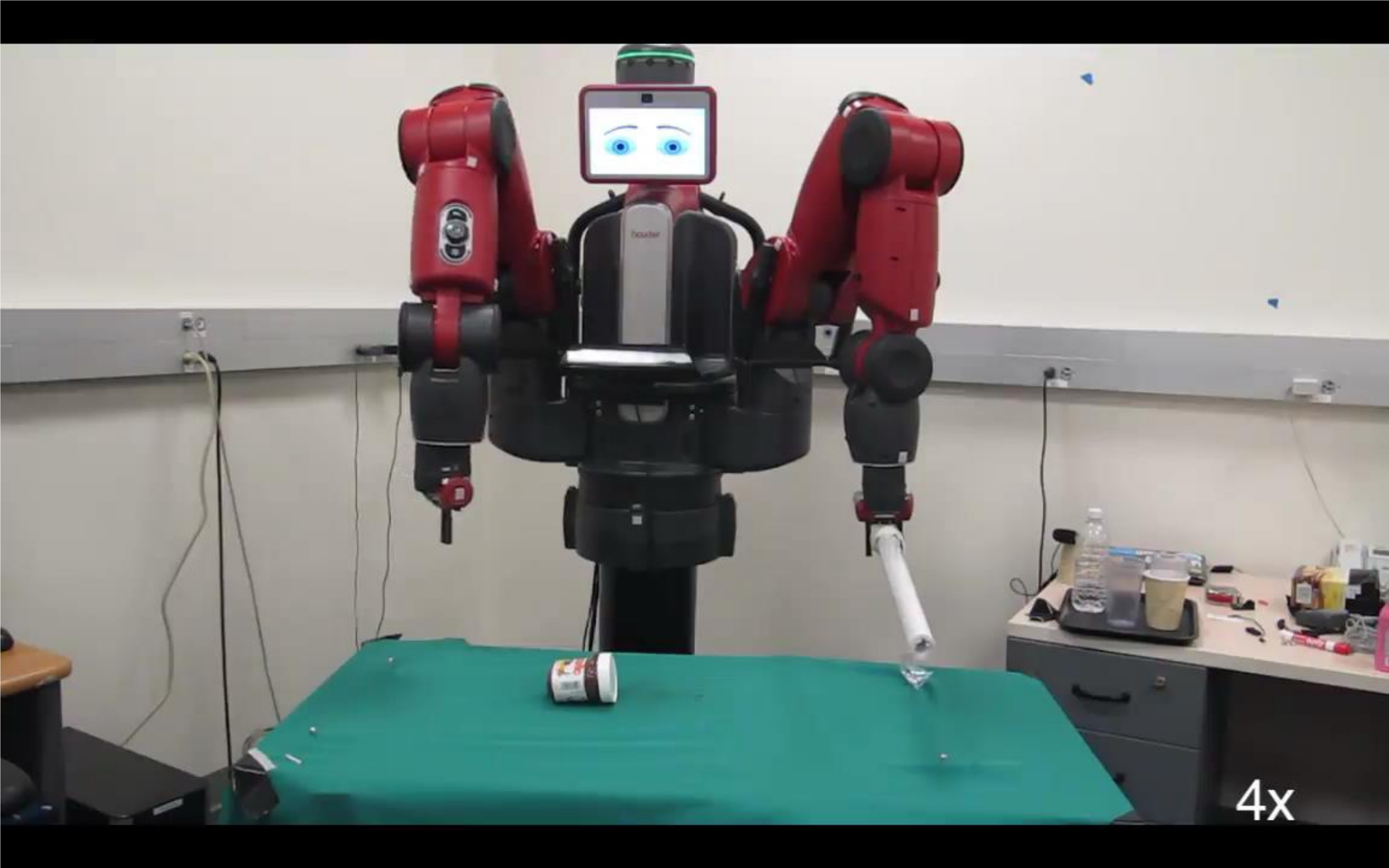
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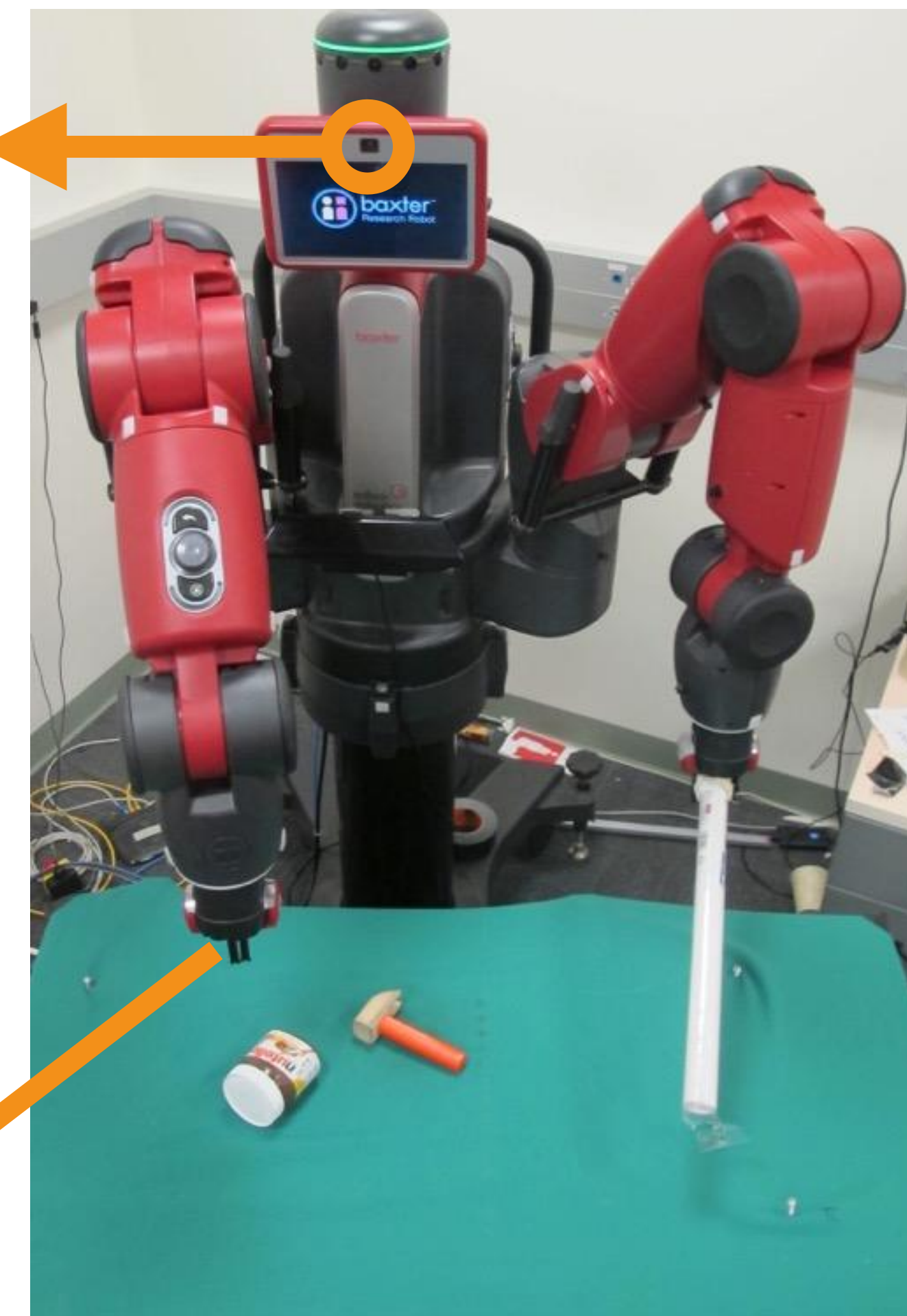
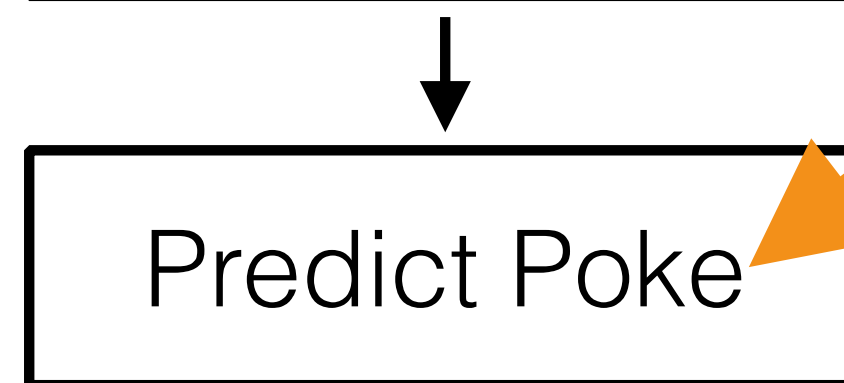
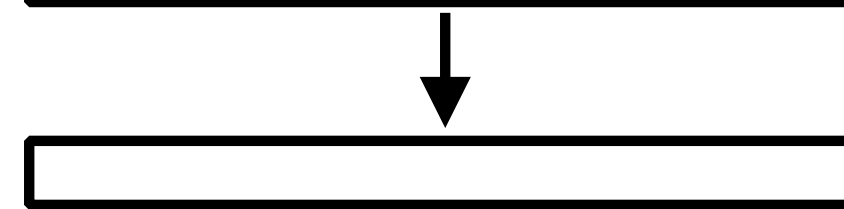
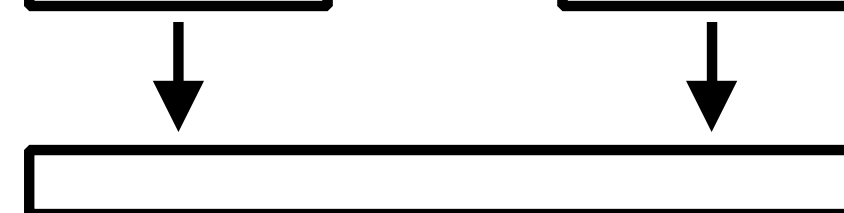
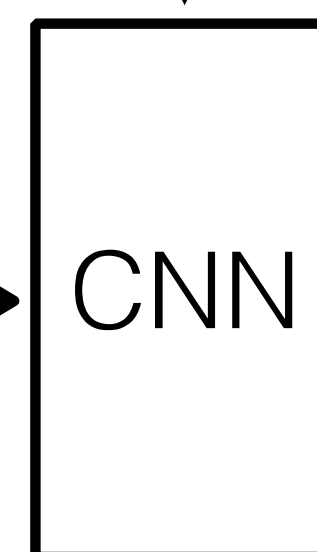
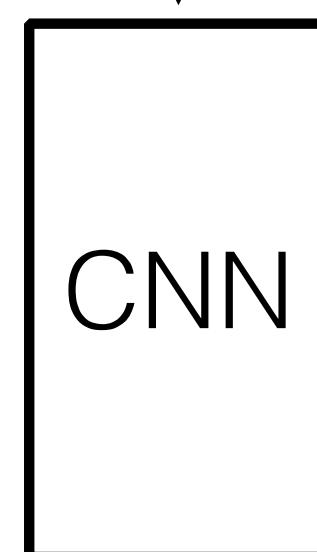
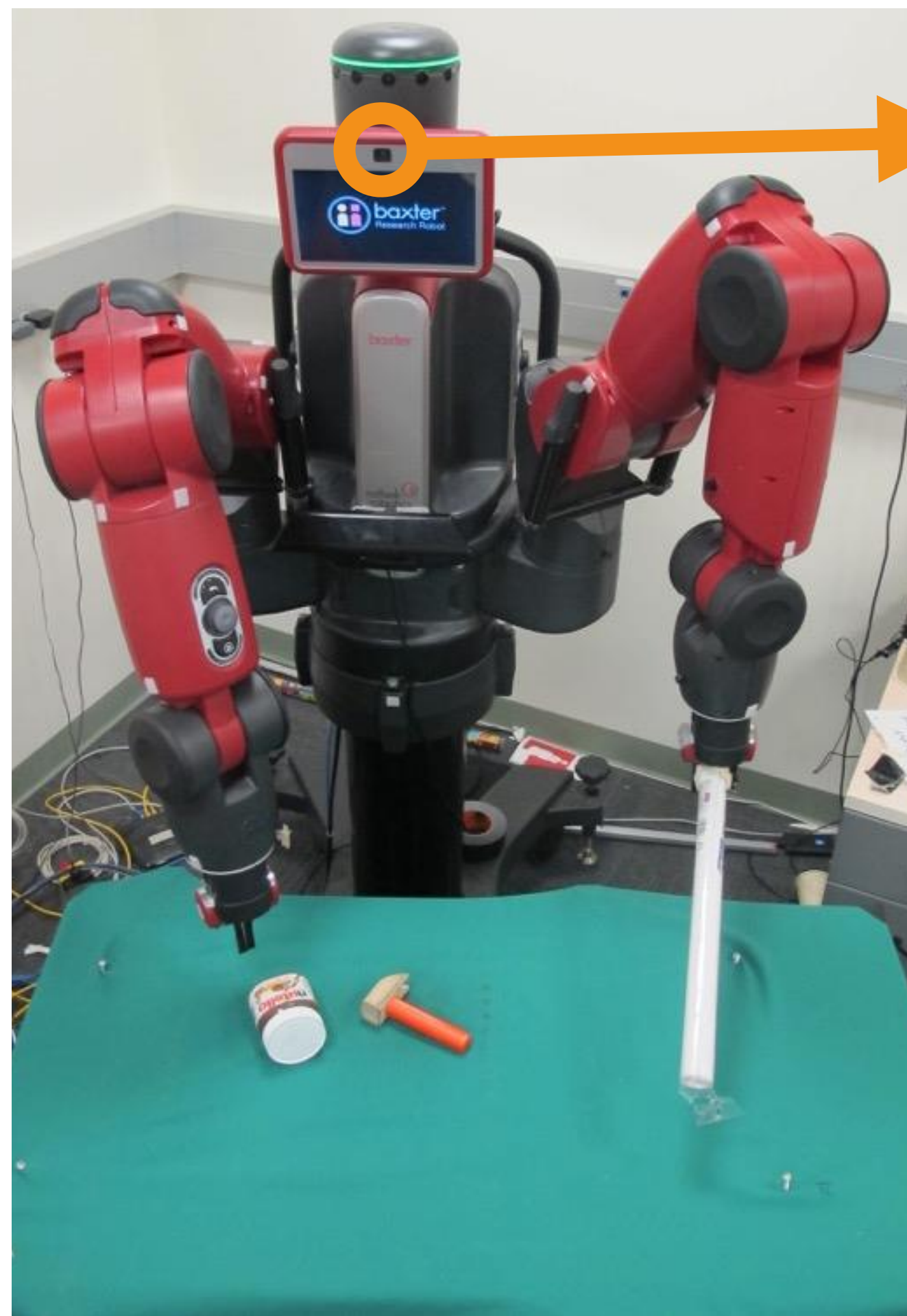
Learning To Poke by Poking: Experiential Learning of Intuitive Physics

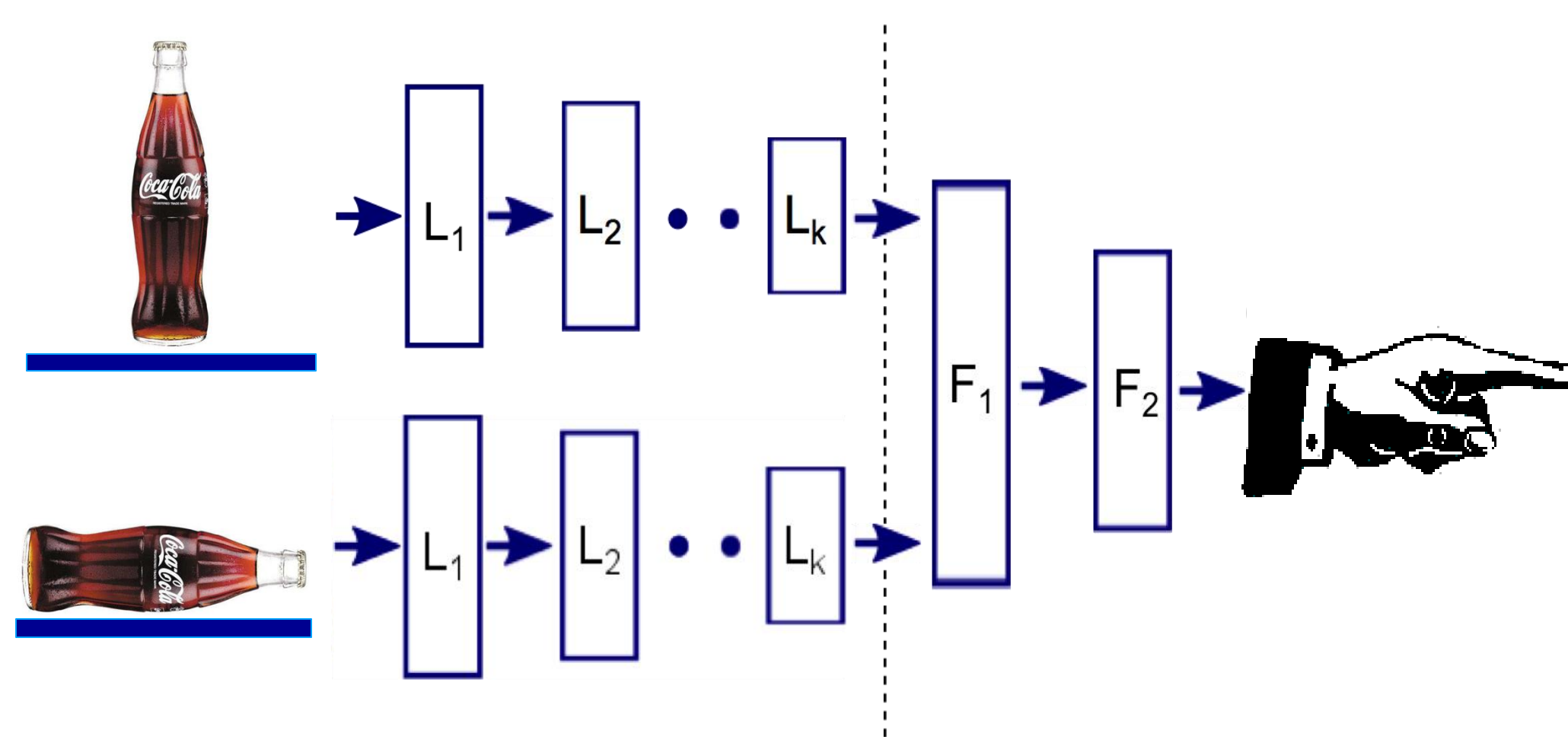
**Pulkit Agrawal*, Ashvin Nair*, Pieter Abbeel, Jitendra Malik,
Sergey Levine**

NIPS 2016

***Equal contribution**

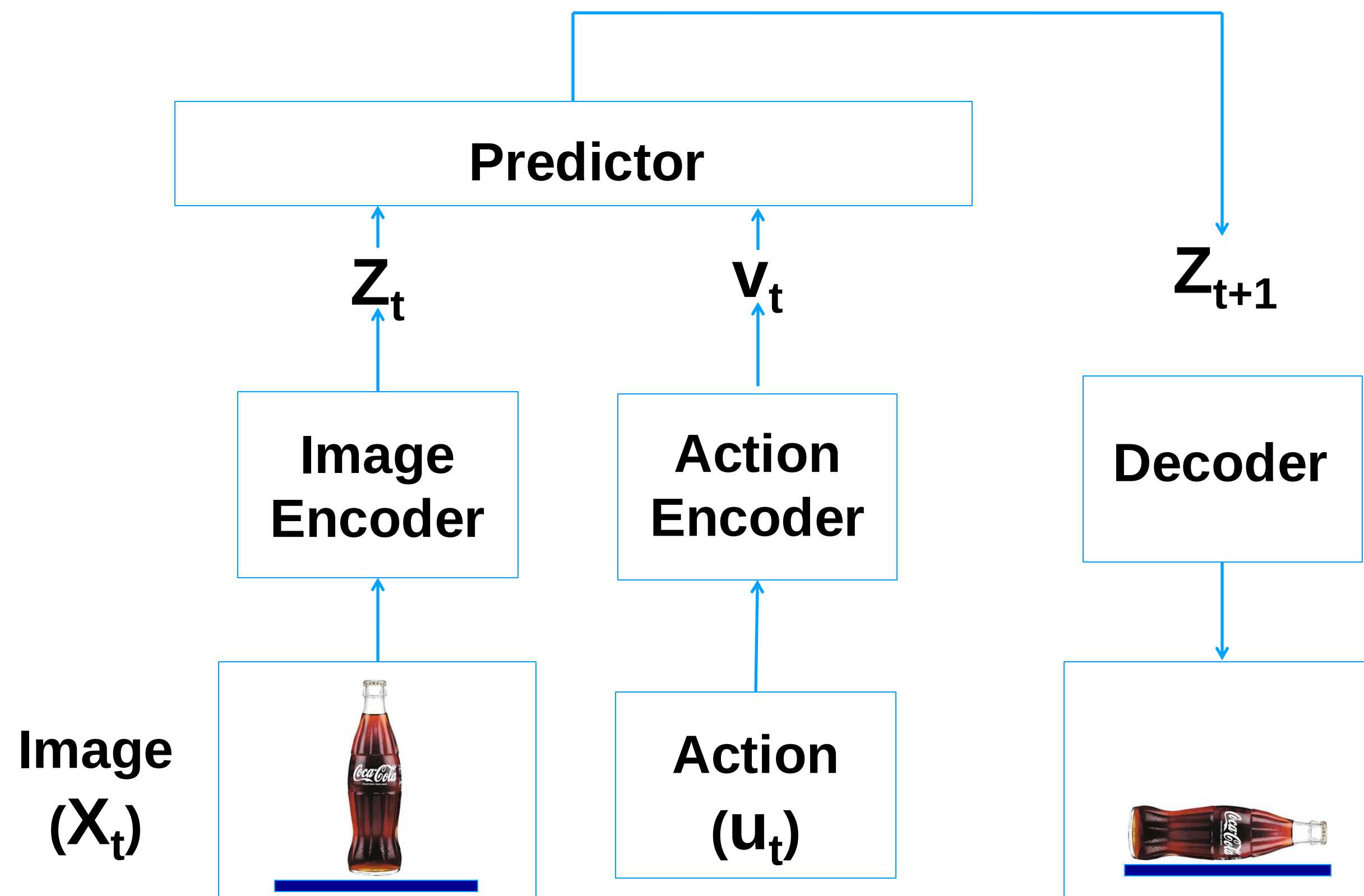






Inverse Model

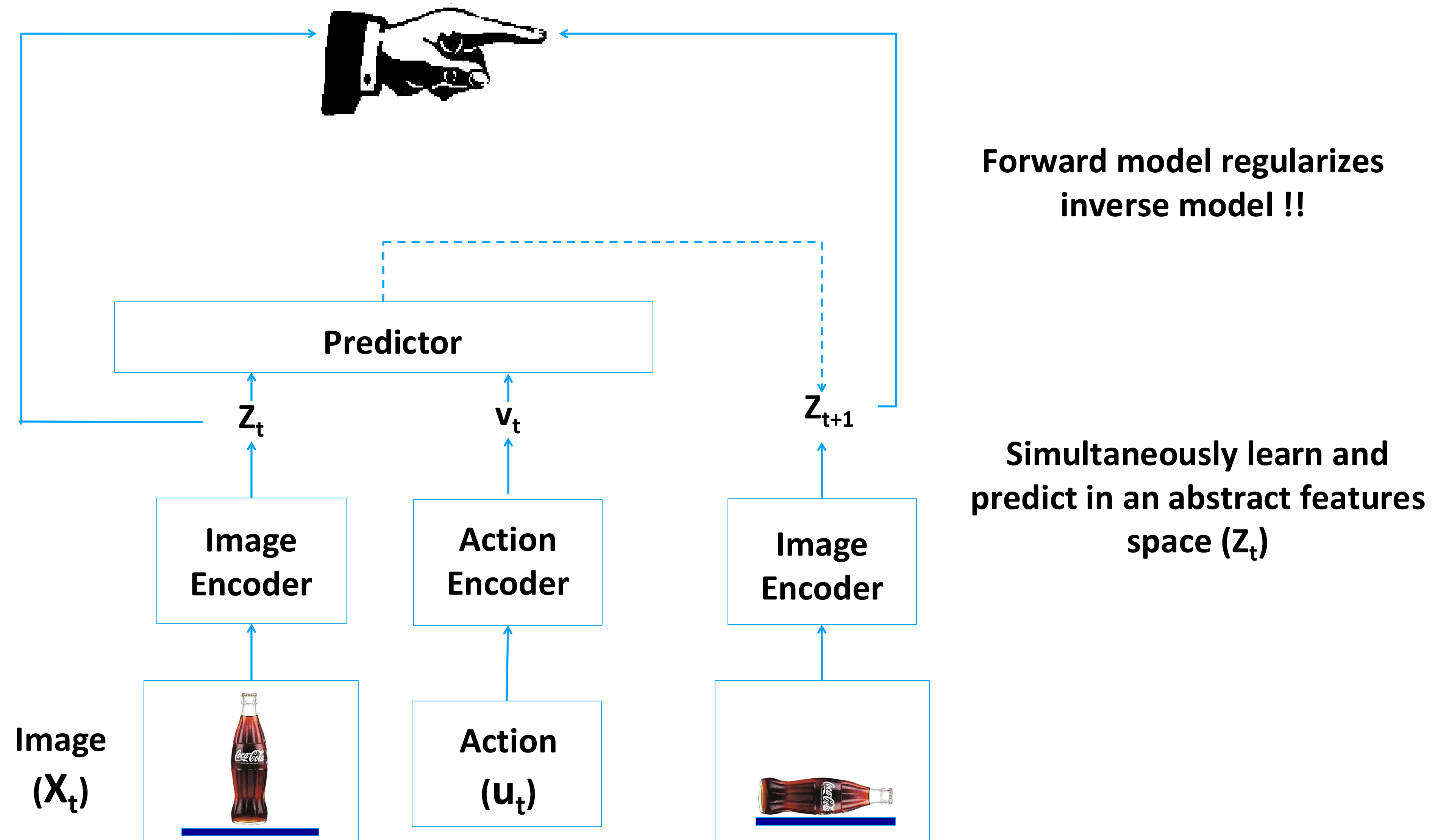
Multimodality in output space



Forward Model

We do not want to predict pixels!!

Our Method: Jointly Learn Forward and Inverse Model

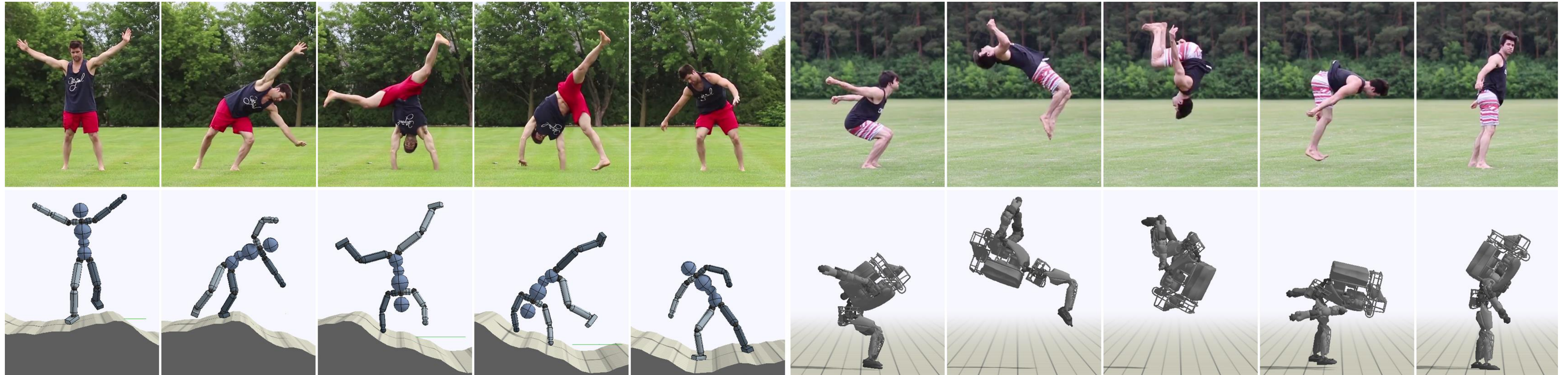


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Learning to act from 4D reconstructions



Peng, Kanazawa, Malik, Abbeel, Levine
Siggraph Asia 2018

The importance of imitation learning has long been recognized in child development; some neuroscientists have argued that “mirror neurons” maybe a mechanism.

Humanoid Locomotion as Next Token Prediction

Ilija Radosavovic Bike Zhang Baifeng Shi Jathushan Rajasegaran Sarthak Kamat
Trevor Darrell Koushil Sreenath Jitendra Malik

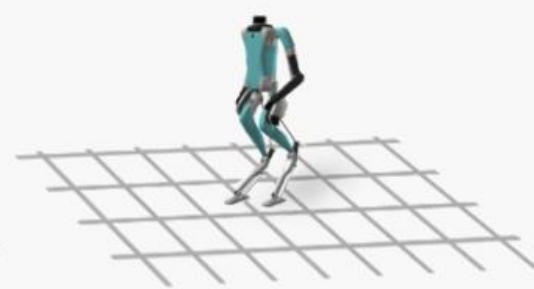
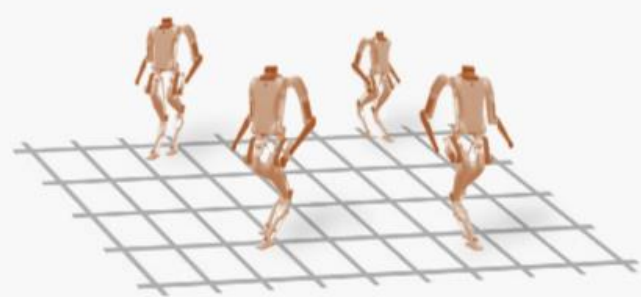
University of California, Berkeley

Humanoid Control as Next Token Prediction

Data

Neural network policy

Model based controller

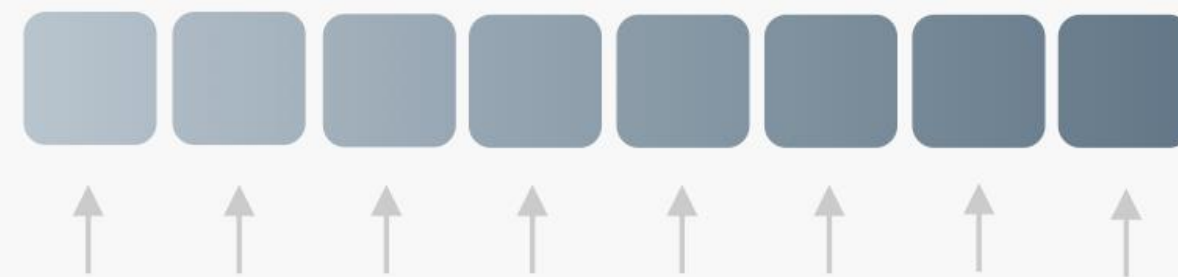


Motion capture

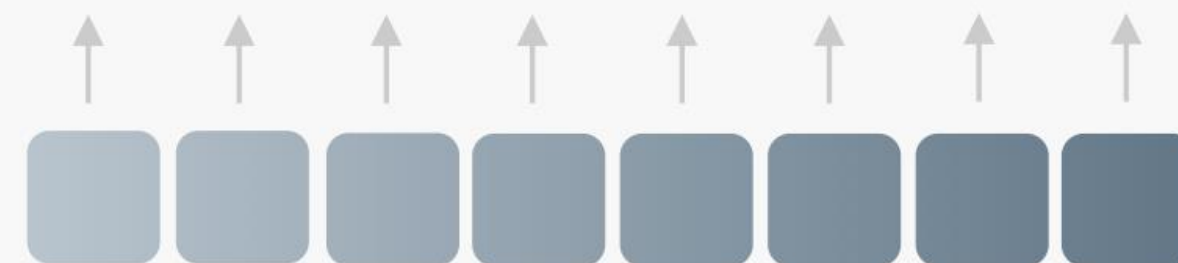
Internet videos



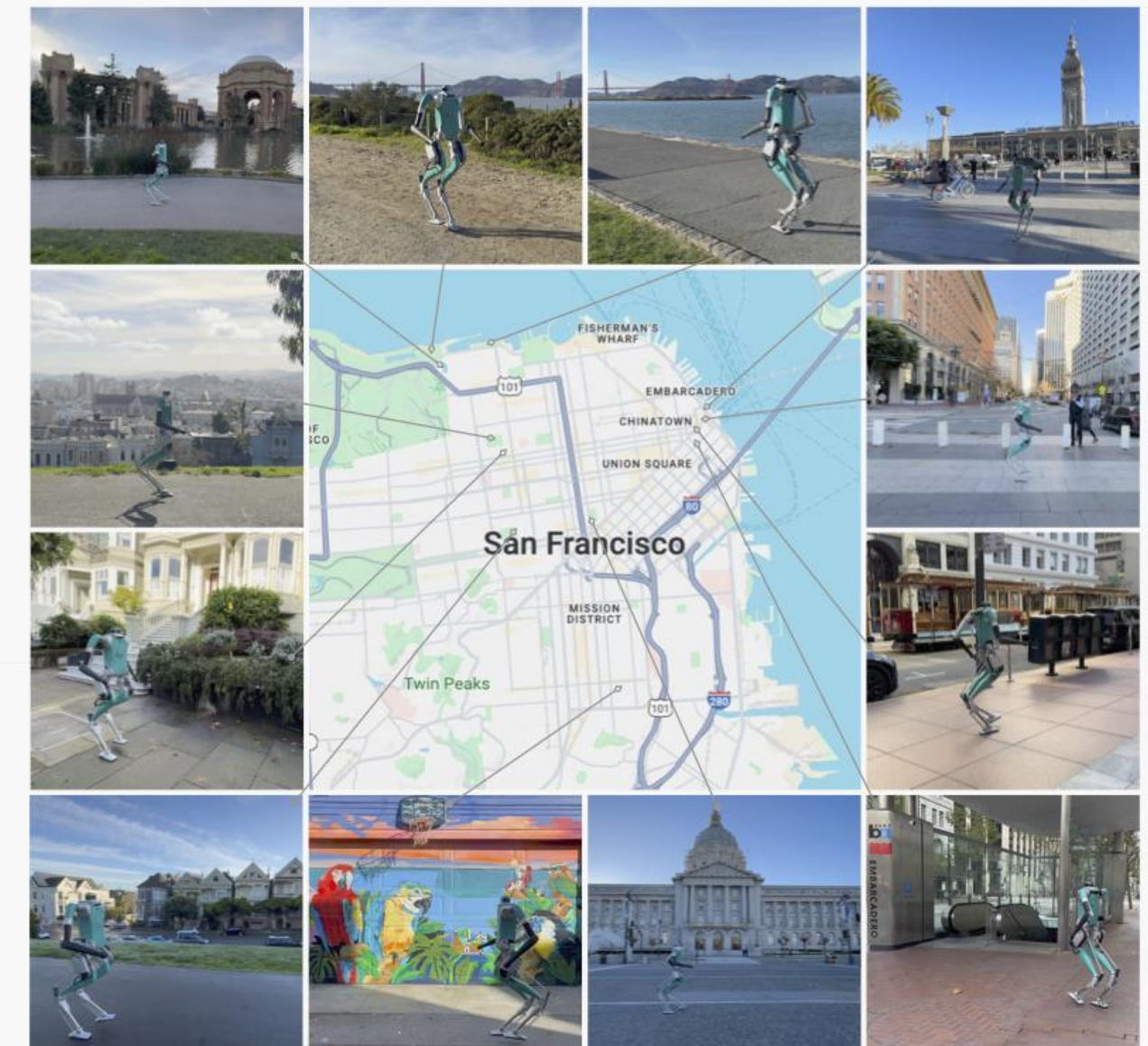
Training



Transformer

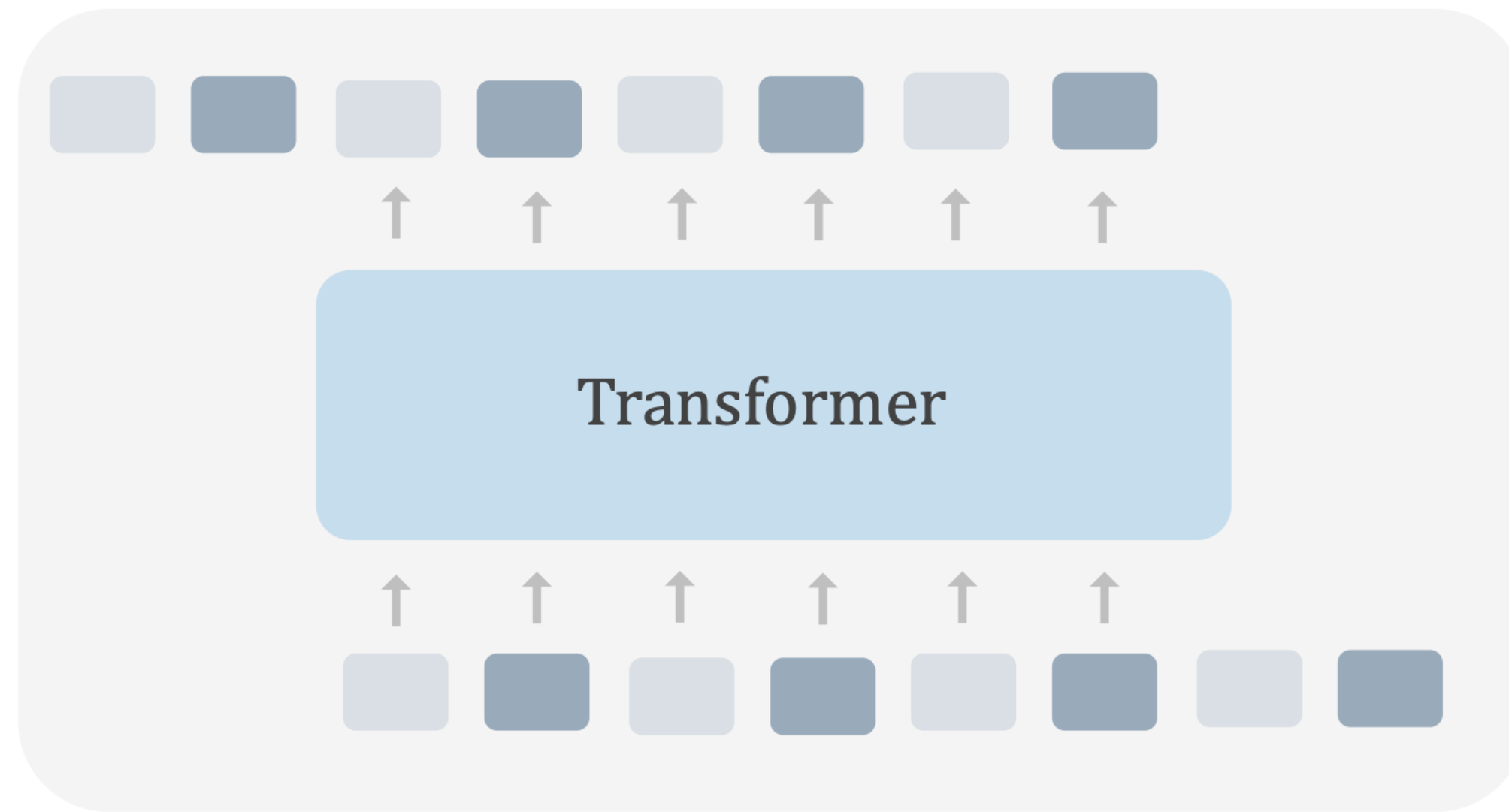


Deployment

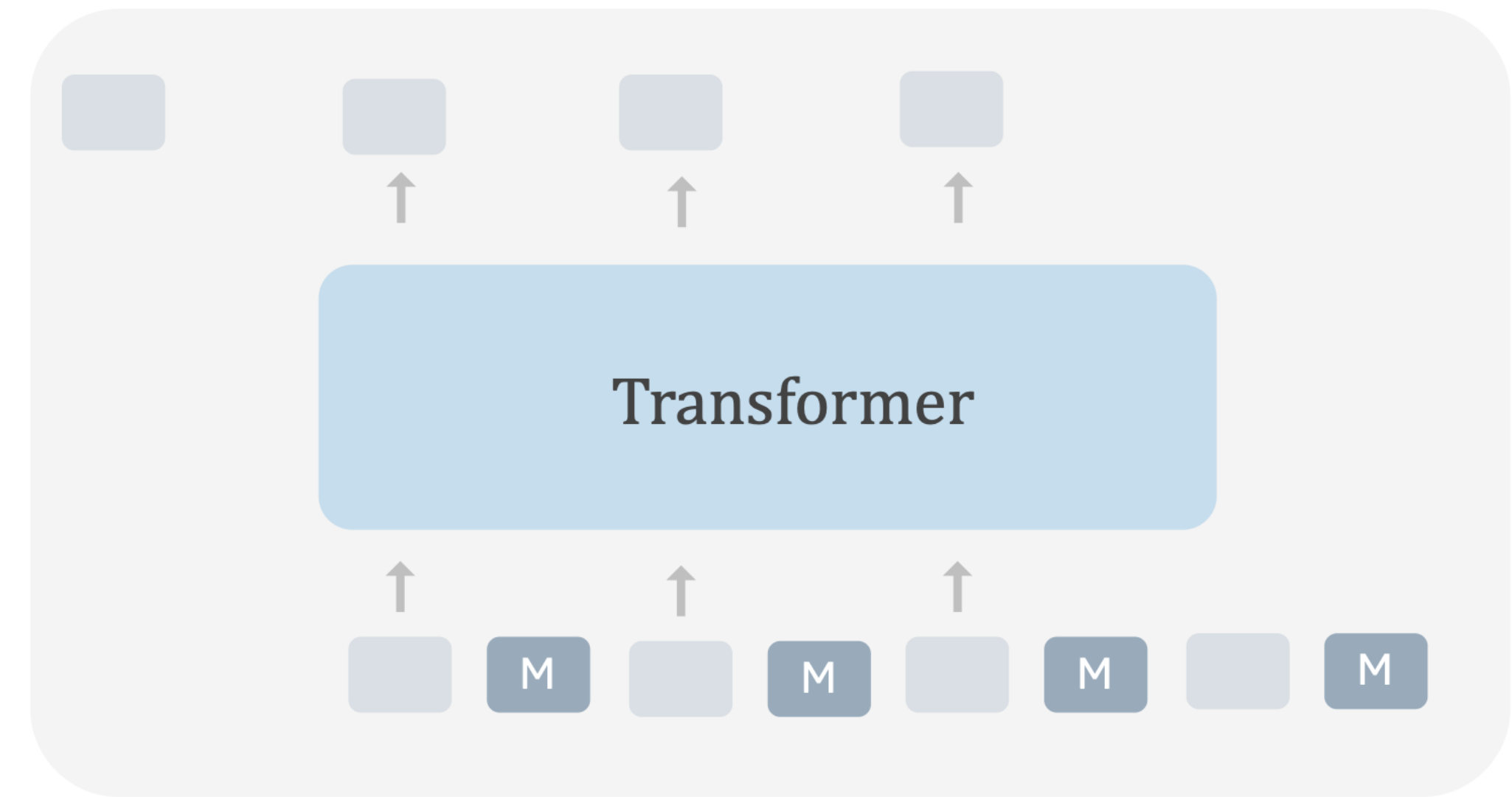


Training with Missing Data

Training with complete data

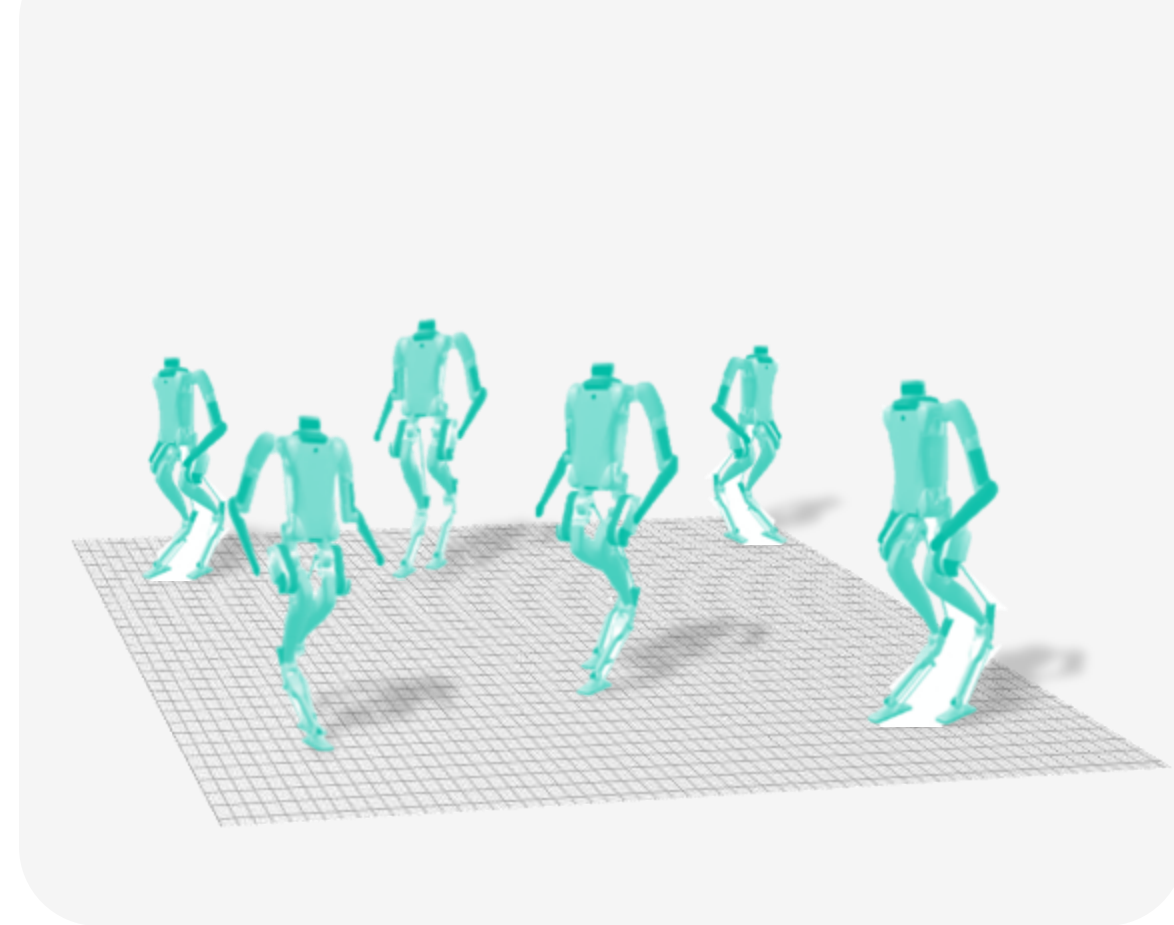


Training with missing data

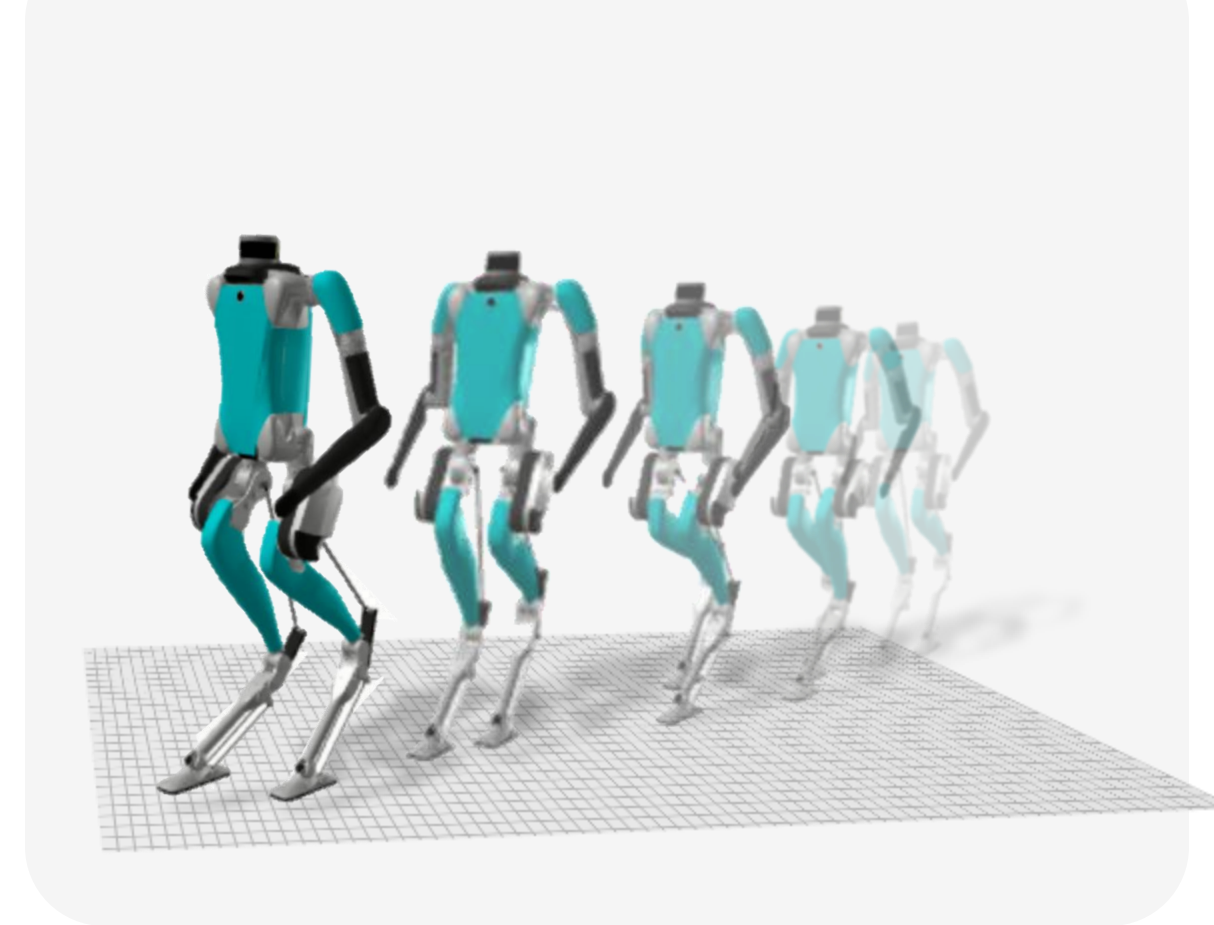


A Dataset of Walking Trajectories

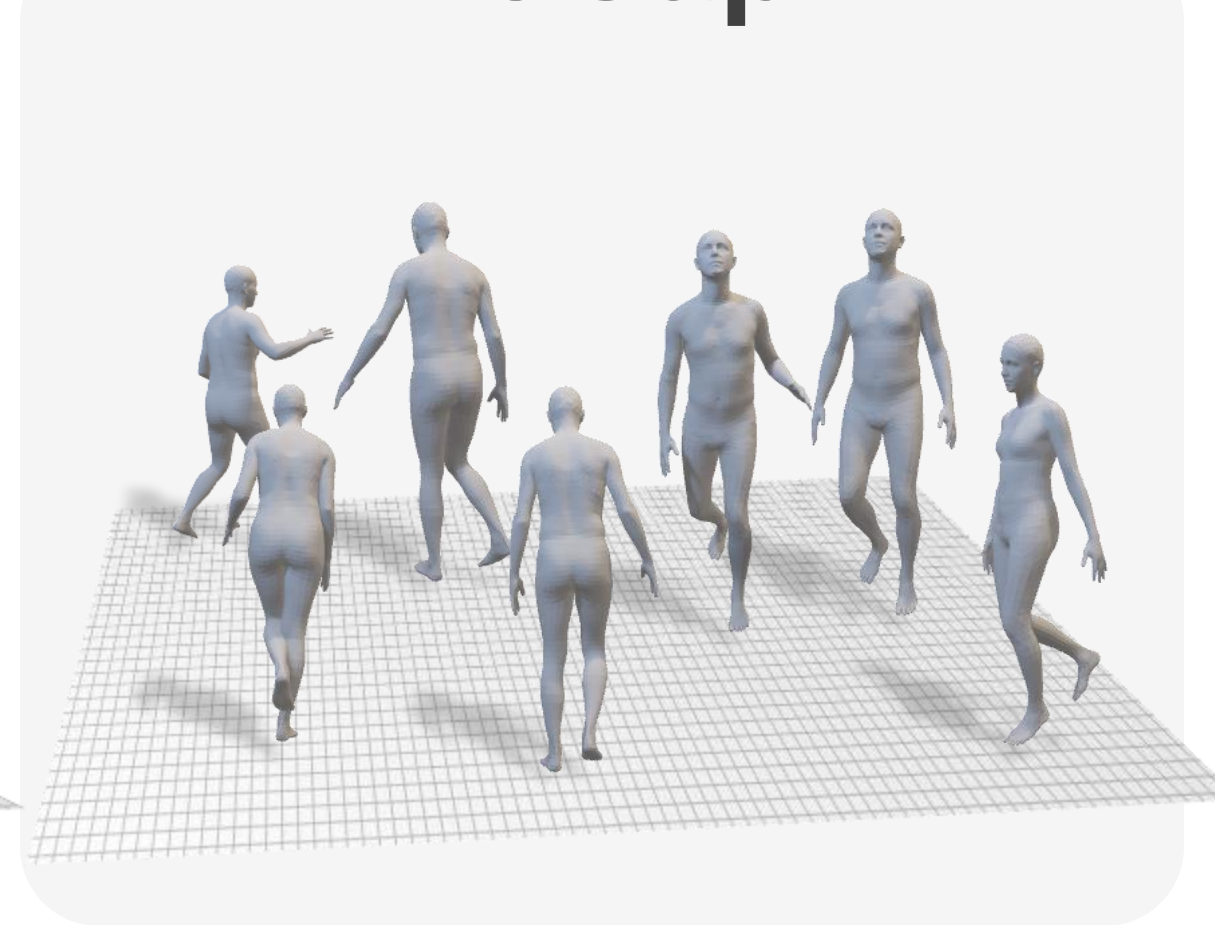
Neural Net Controller Model based Controller



Model based Controller MoCap



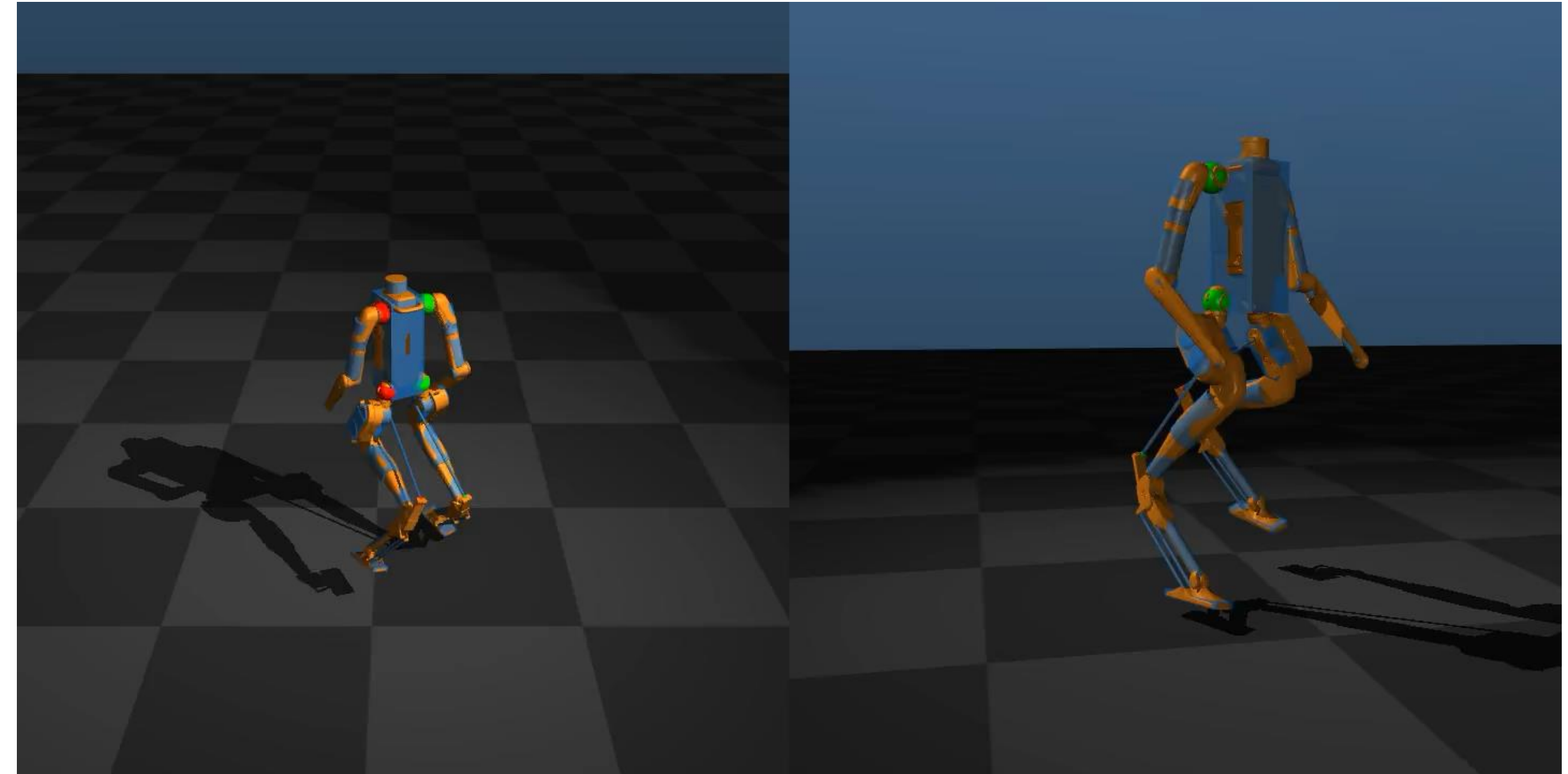
MoCap



Internet Videos



Retarget YouTube Video in Simulation

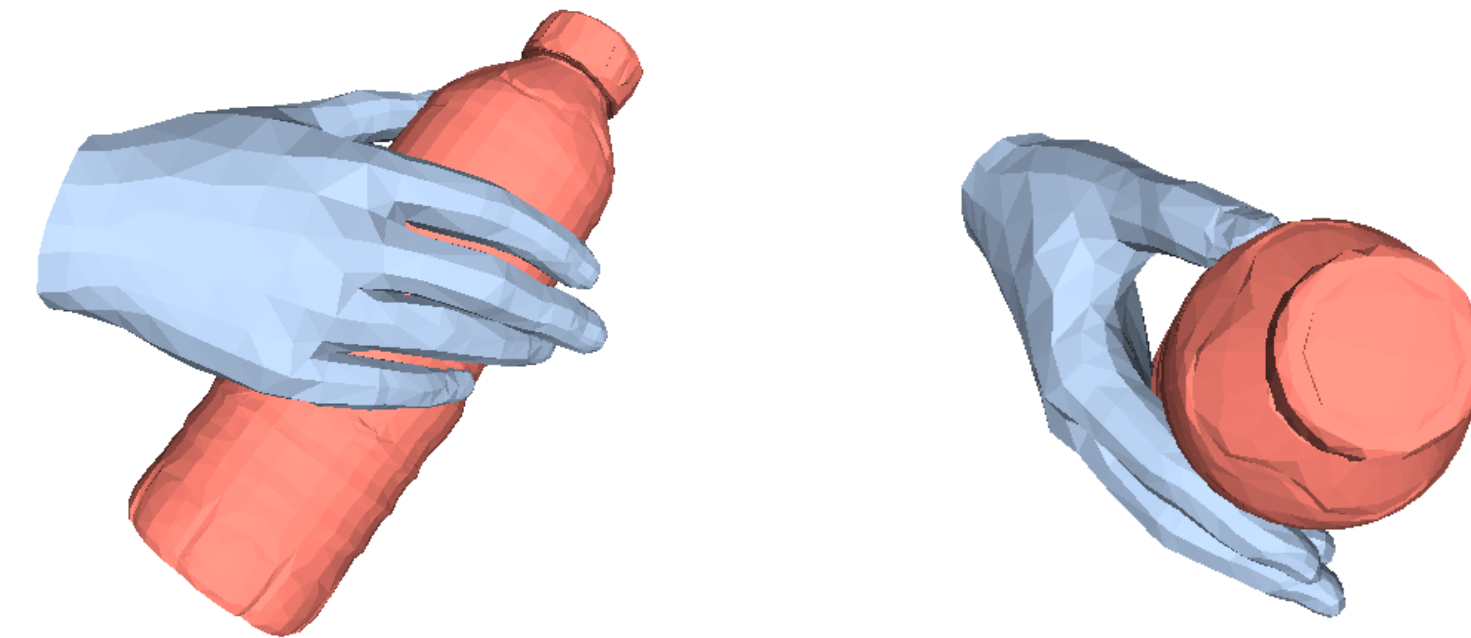




Imitation Learning for Robot Manipulation



image



3D reconstruction

Reconstructing Hand-Object Interactions in the Wild
Zhe Cao, Ilija Radosavovic, Angjoo Kanazawa, Jitendra Malik
(ICCV 2021)



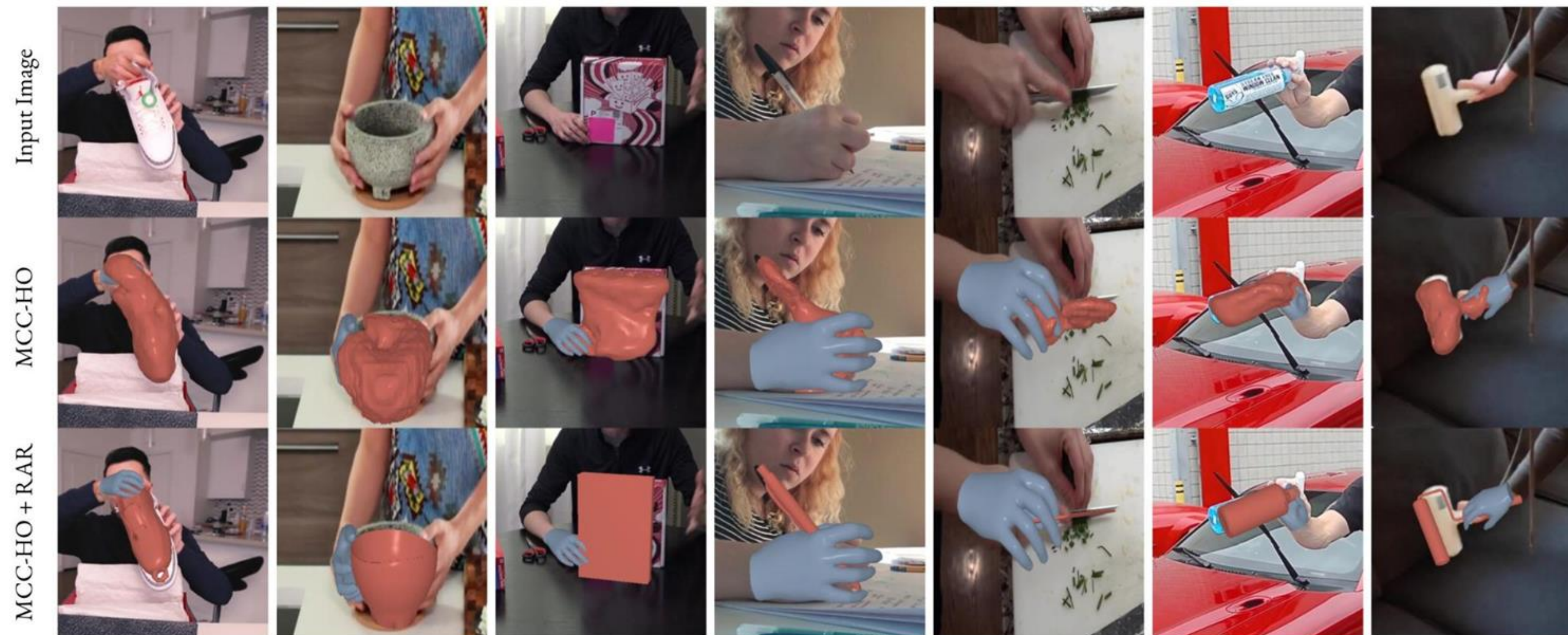
Reconstructing Hands in 3D with Transformers

Georgios Pavlakos, Dandan Shan, Ilija Radosavovic, Angjoo Kanazawa, David Fouhey, Jitendra Malik

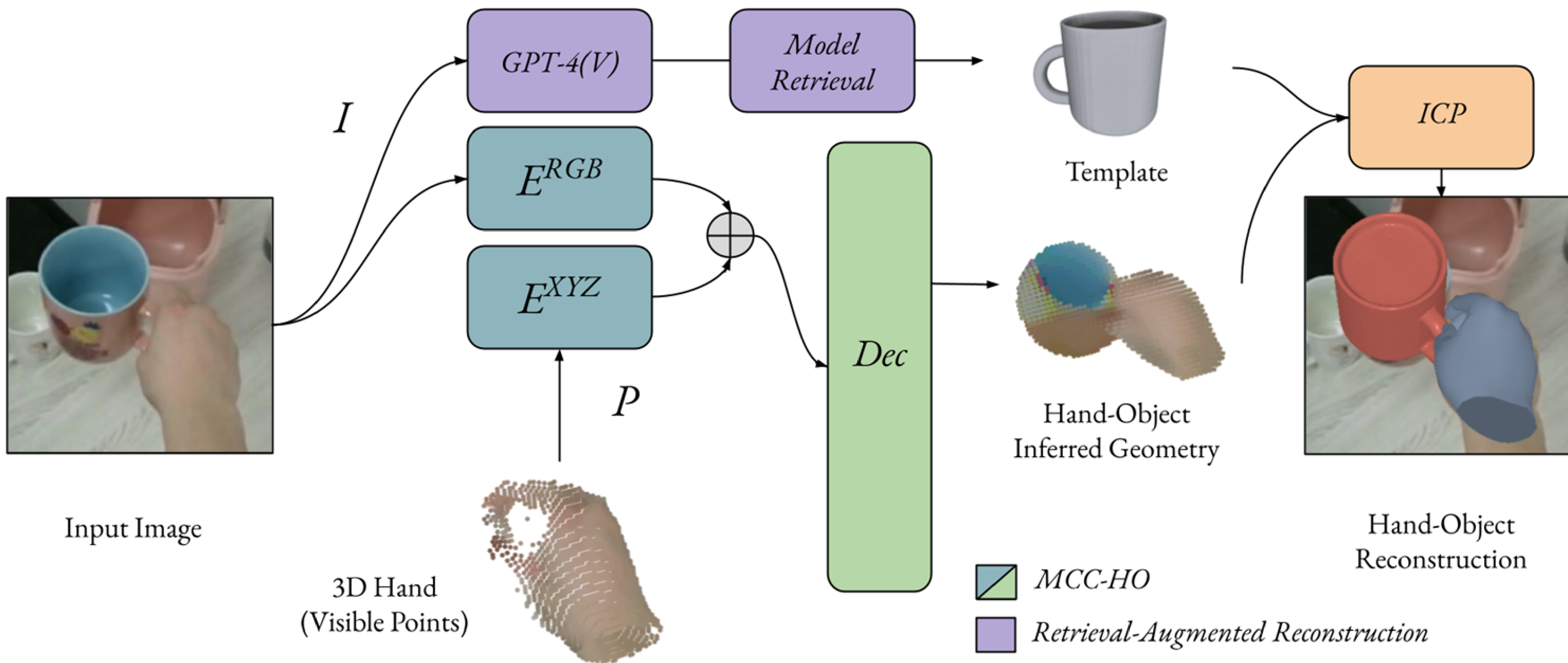
Reconstructing Hand-Held Objects in 3D

Jane Wu, Georgios Pavlakos, Georgia Gkioxari,
Jitendra Malik

<https://arxiv.org/abs/2404.06507>



Builds on two previous foundation models : HaMeR (Berkeley) and MCC (Meta)



Reconstruction

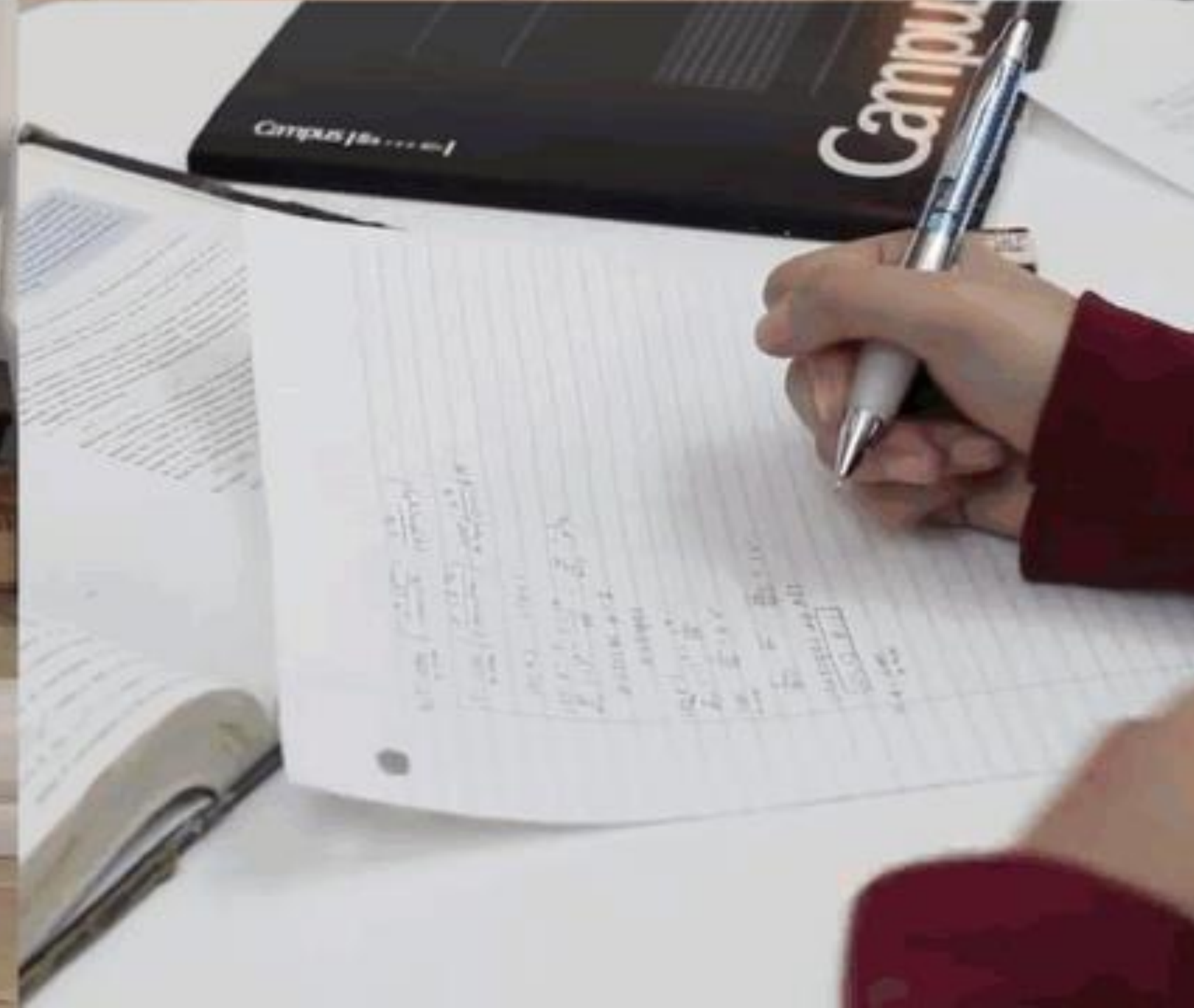
MCC-HO Results



	DexYCB			MOW			HOI4D		
	F-5 (↑)	F-10 (↑)	CD (↓)	F-5 (↑)	F-10 (↑)	CD (↓)	F-5 (↑)	F-10 (↑)	CD (↓)
HO [24]	0.24	0.48	4.76	0.03	0.06	49.8	0.28	0.51	3.86
IHOI [90]	-	-	-	0.13	0.24	23.1	0.42	0.70	2.7
MCC-HO	0.36	0.60	3.74	0.15	0.31	15.2	0.52	0.78	1.36

Comparison to prior work. Chamfer Distance (cm²) and F-score (5mm, 10mm).





Thank you!